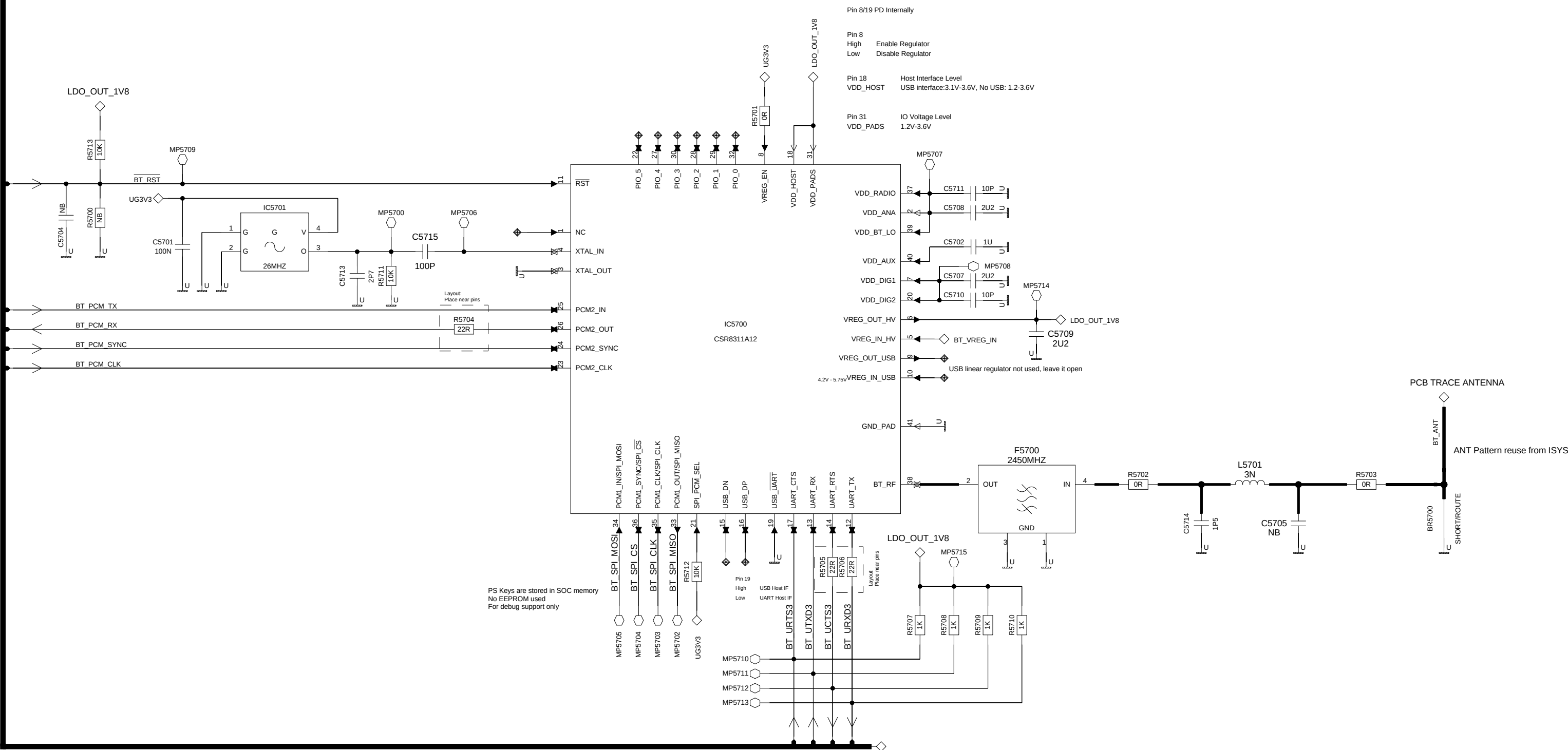
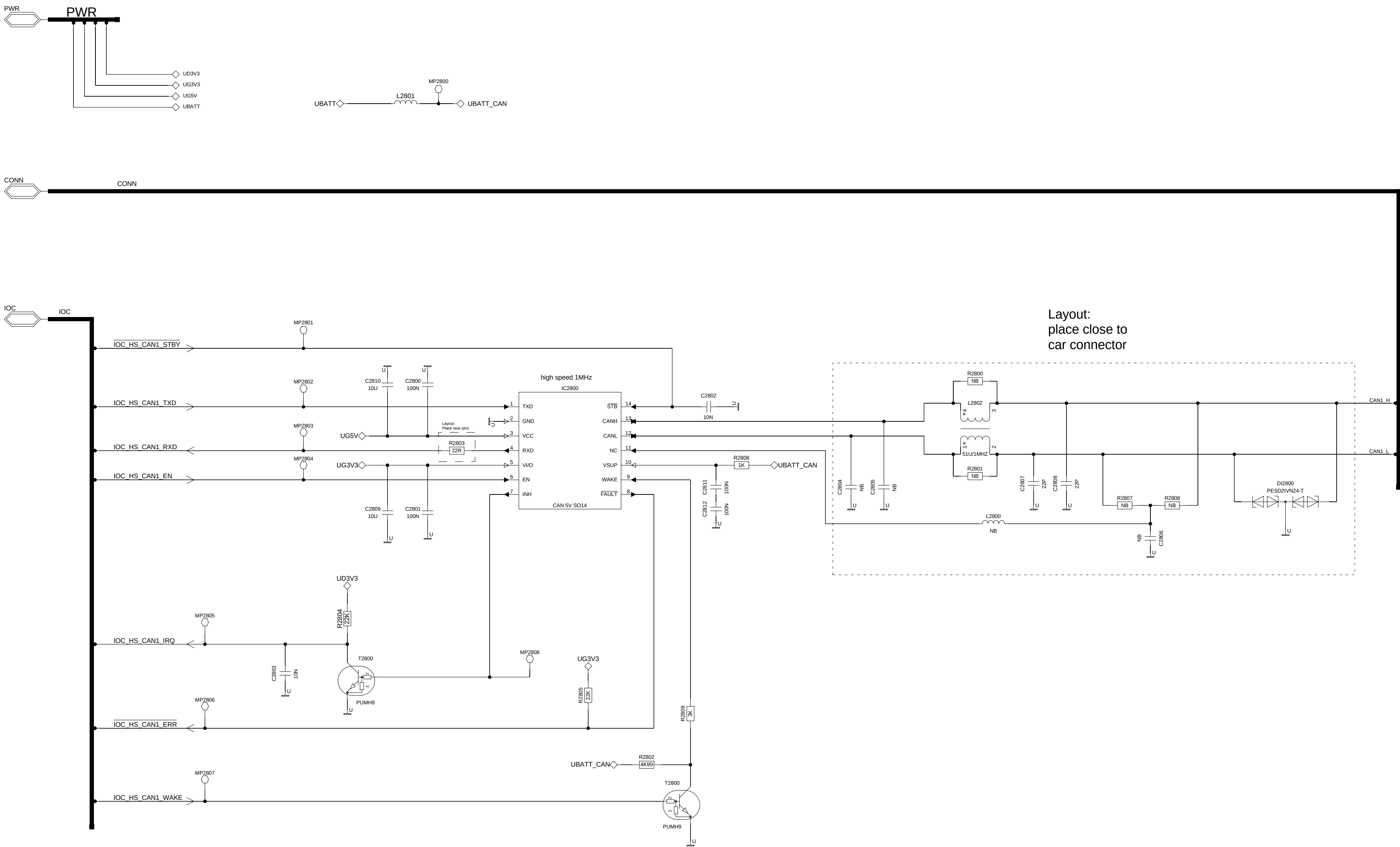


The schematic shows the power supply network for the BT module. It includes a power source labeled "PWR" connected to a series combination of a 600ohm resistor and a 0.165ohm DCR component at 500mA. This network is connected to the BT_VREG_IN pin of the MP5701 regulator. The regulator's output is connected to the BT module. A 22uF capacitor (C5703) is connected between the input and output of the regulator. Other components shown include a 2u2 capacitor (C5706), a 100N capacitor (C5700), and a .00P capacitor (C5712).



Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 2	Sheets of 31	Sample
4010_Bluetooth			Type	Variant	Part Number
substitute for:		substitute by:		C143	

CAN (2800-2899)

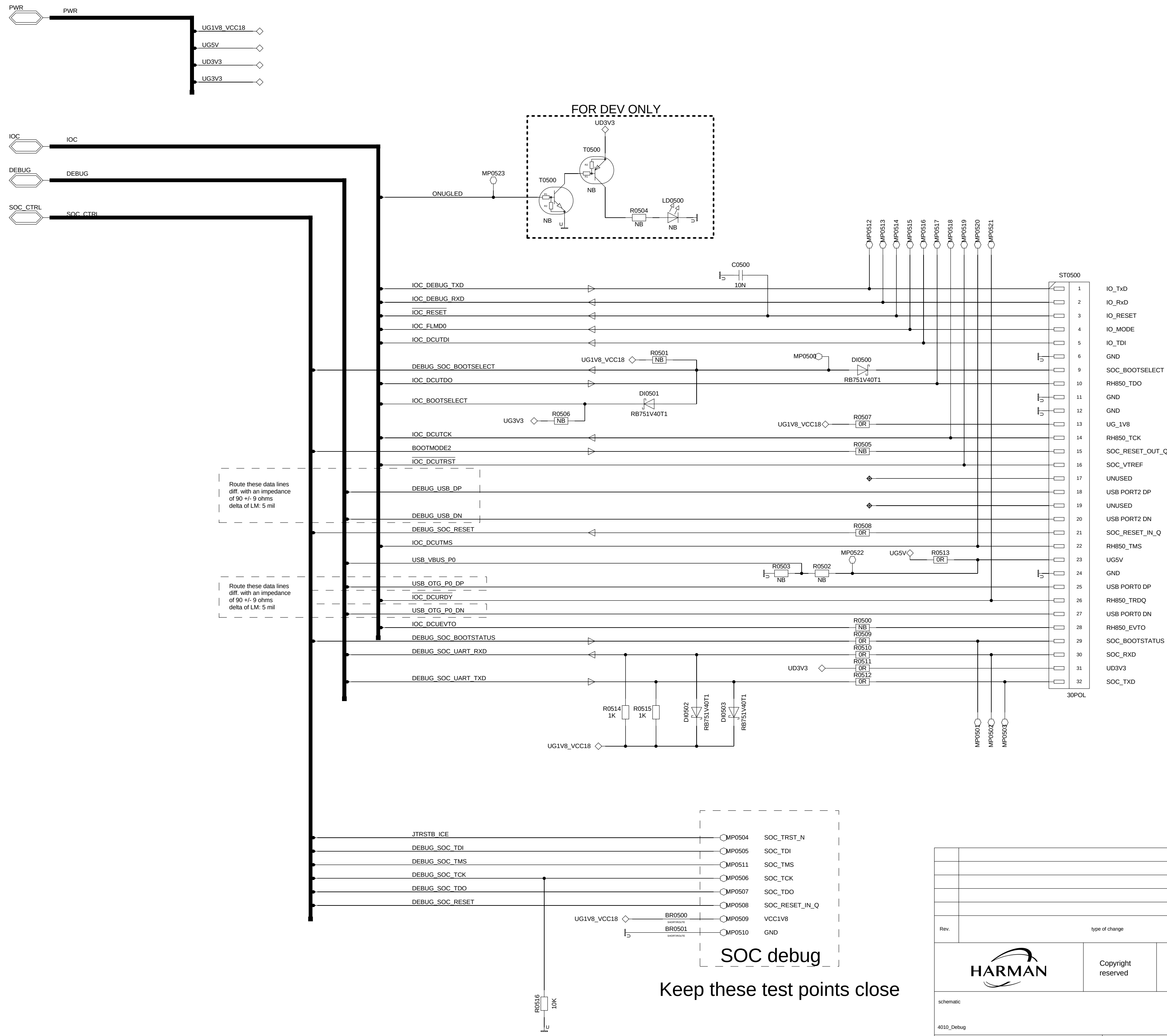


Rev.	type of change		change-no.	date of change	auth.	checked		
		Copyright reserved	Wiring modifications reserved					
					date		name	
					created			
			checked					
schematic				Sheet	Sheets of	Sample		
4010_CAN				3	31			
substitute for:				Type	Variant	Part Number		
substitute by:								

The schematic diagram illustrates the ATMX1057T-A19 board, centered around the IC7000. The board includes several key components and connections:

- Power and Grounding:** The board is powered by a 3.3V supply (UG3V3_L7000) and a 5V supply (UG3V3_L7001). Grounding is established through various pins (NC_1 to NC_60) and a dedicated ground plane (GNDUP).
- IC7000 Connections:** The IC7000 is connected to various pins, including TP_RST, TP_RSTQ, TP_SCL, TP_SDA, TP_SYNC, and TP_TsRx. It also interfaces with the ATMX1057T-A19 via pins NC_1 to NC_60.
- Memory and Storage:** The board includes a 16MB memory module (MP7000) and a 16MB storage module (MP7005).
- Communication and Control:** The board features a USB-to-UART bridge (MP7001), a USB-to-serial bridge (MP7002), and a USB-to-serial bridge (MP7003). It also includes a USB-to-serial bridge (MP7004) and a USB-to-serial bridge (MP7005).
- Other Components:** The board includes a 16MB memory module (MP7000), a 16MB storage module (MP7005), and a 16MB memory module (MP7001).

The diagram shows the detailed wiring and component placement for the ATMX1057T-A19 board, providing a comprehensive view of its internal architecture and external interfaces.



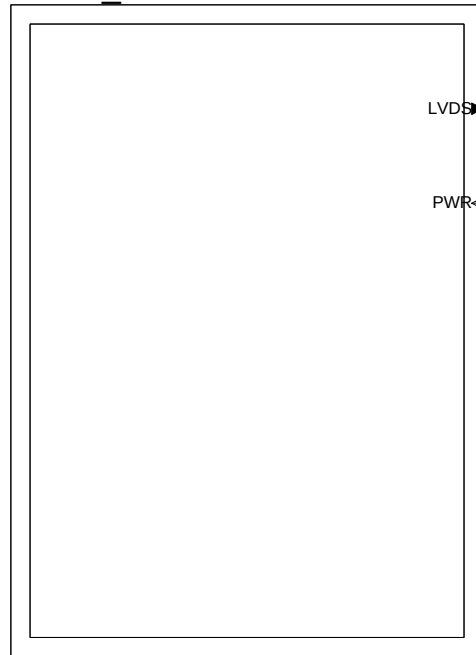
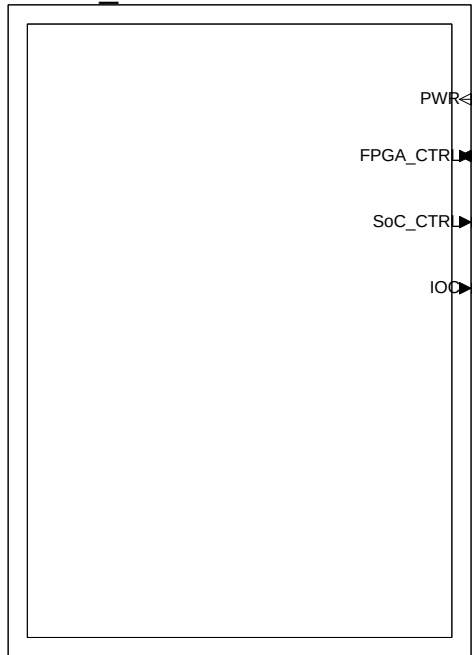
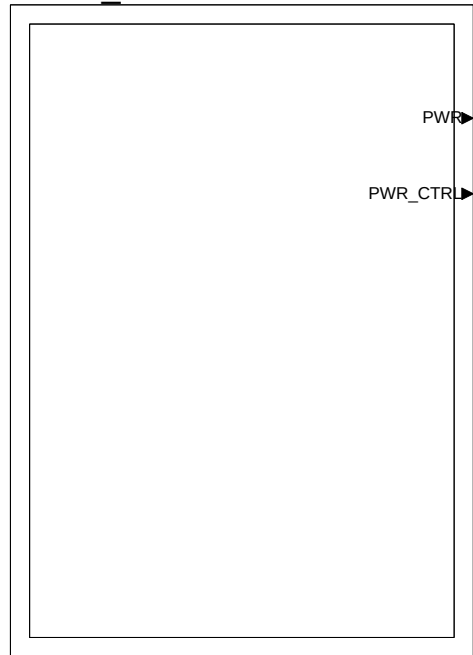
Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 5	Sheets of 31	Sample
4010_Debug			Type	Variant	Part Number
substitute for:		substitute by:			

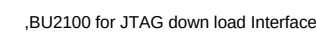
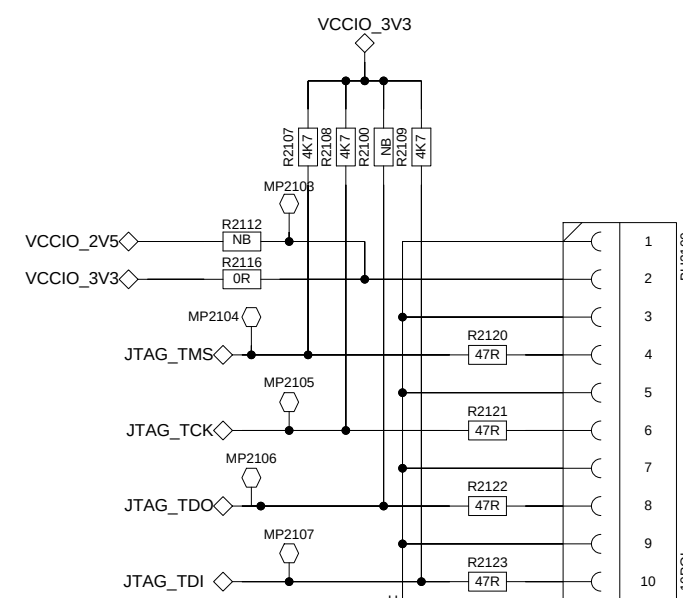
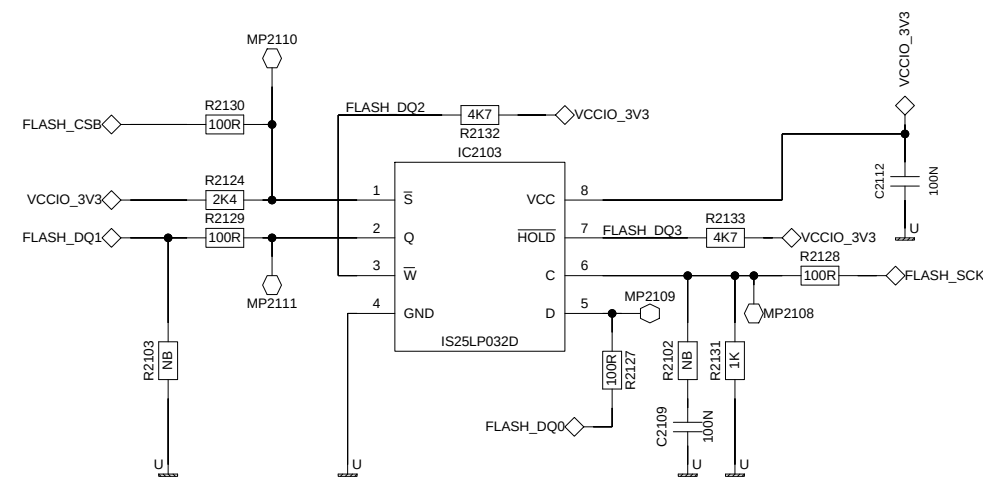
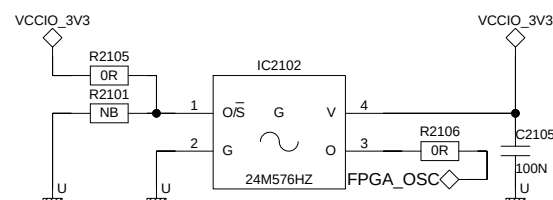
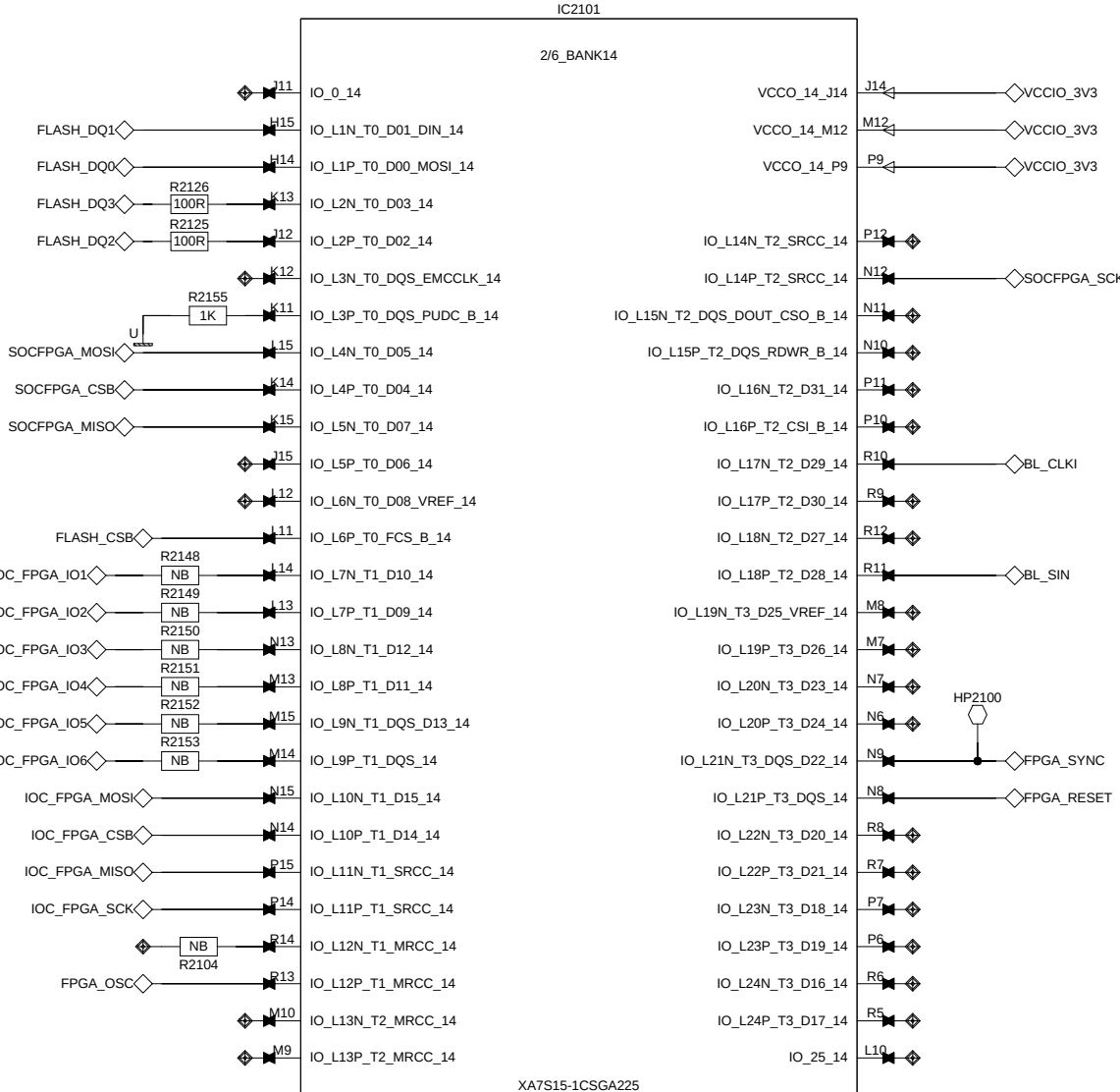
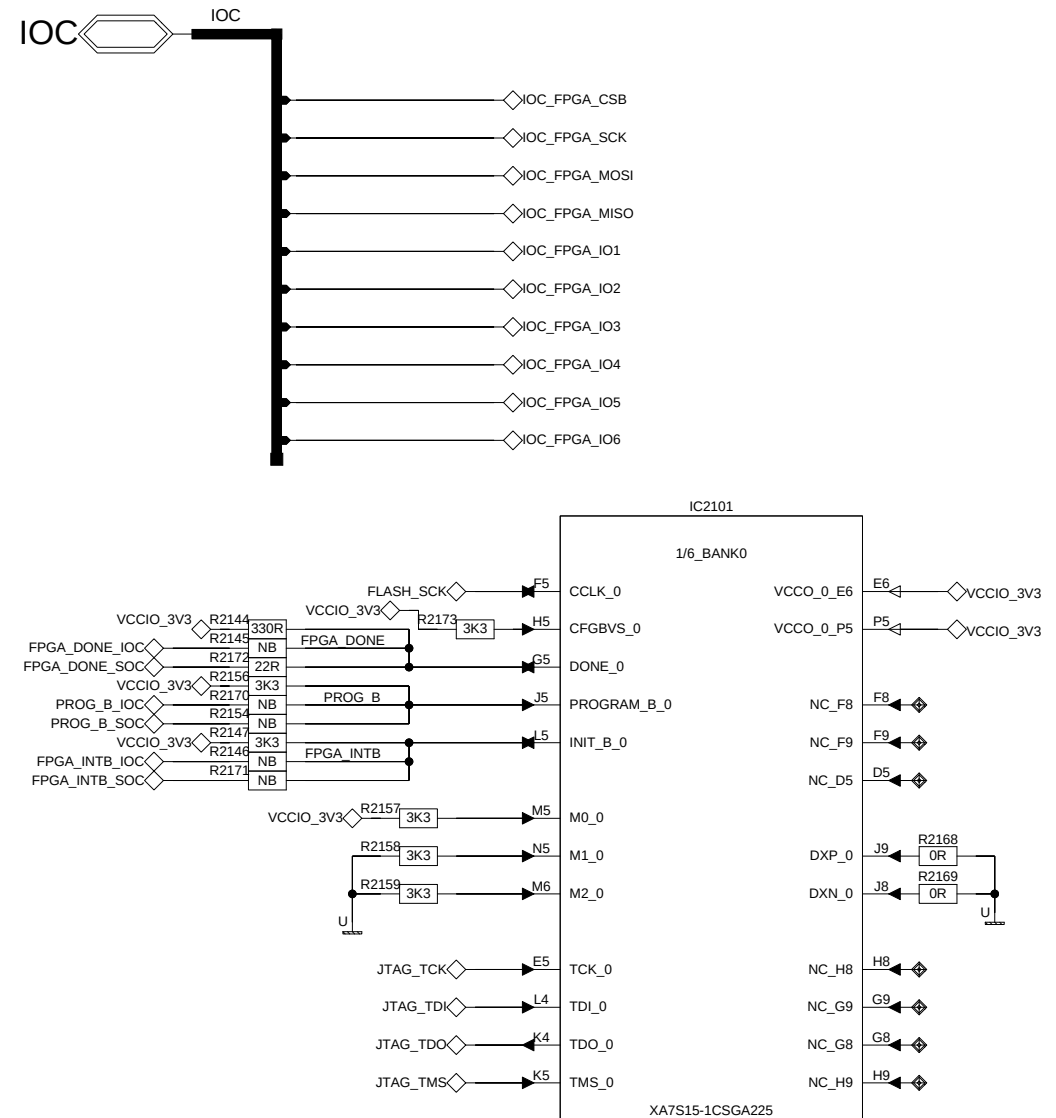
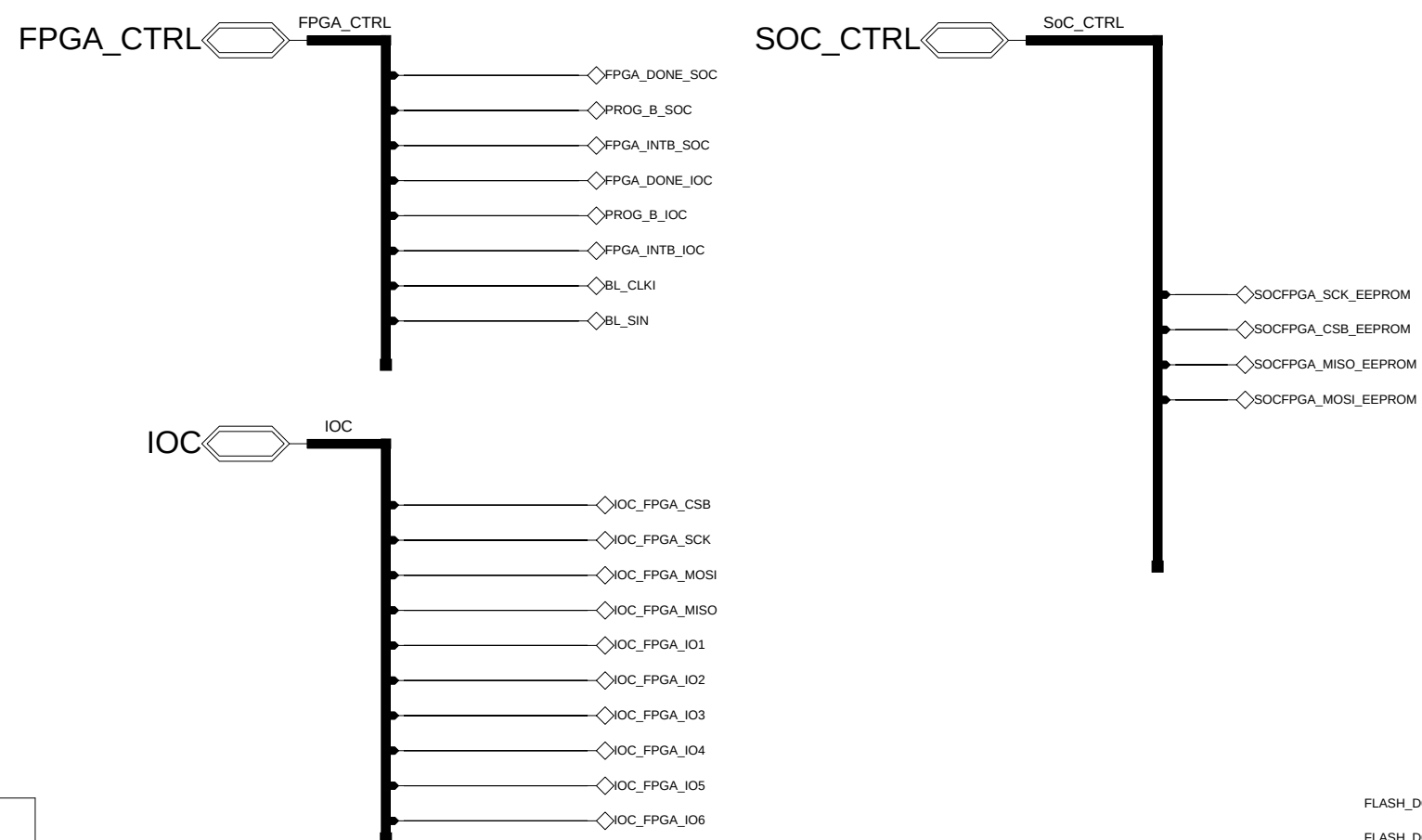
Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

The four circuit diagrams illustrate the power and ground connections for the 1V8_1010PHY component. Each diagram shows a network of components connected to a common ground plane (labeled 'U').

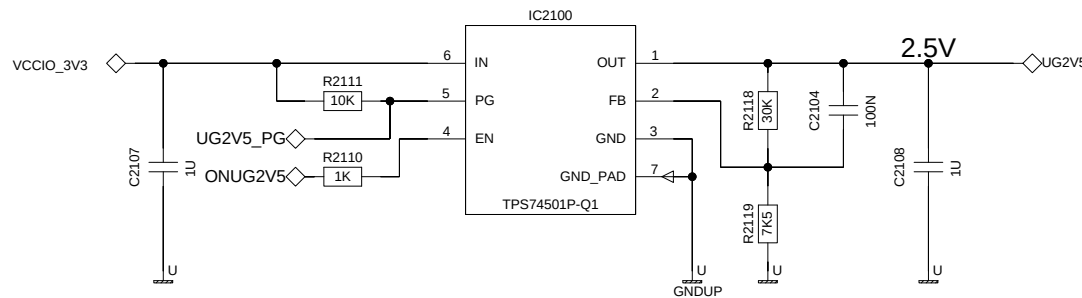
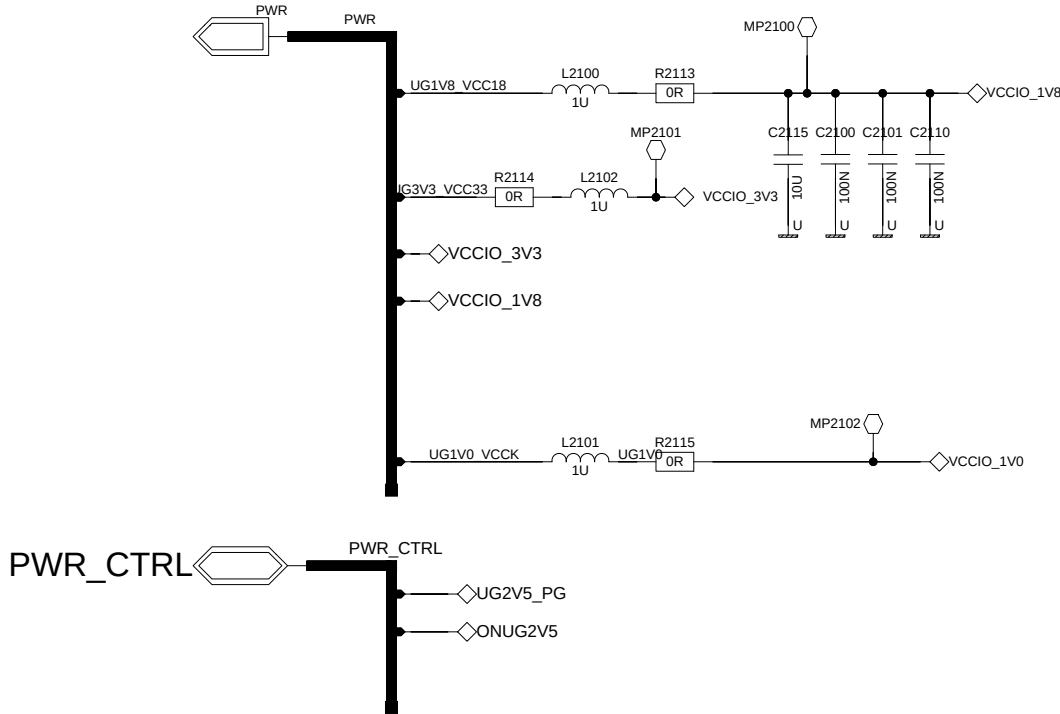
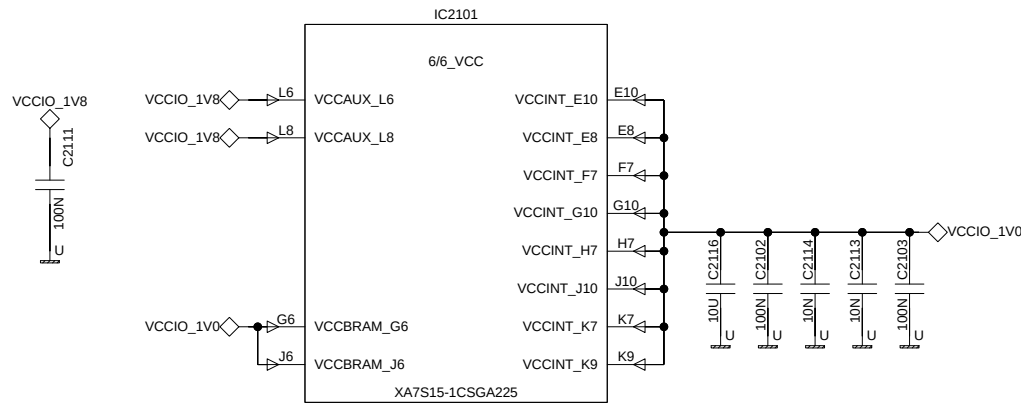
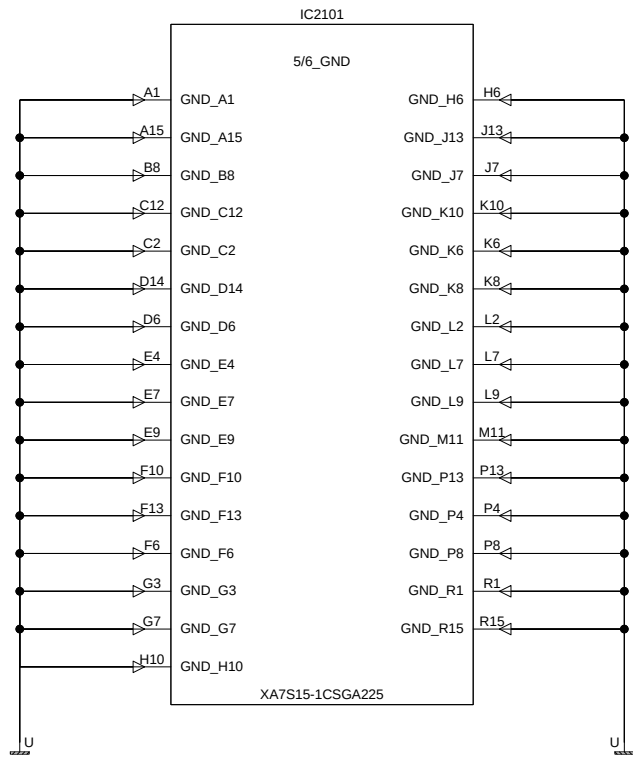
- Top Left Diagram:** Shows the connection of UG3V3, UG3V3_ETH_1010, and MP8301. The network includes inductor L8300, capacitors C8312 (22uF), C8305 (1uF), C8306 (1uF), C8309 (1uF), and a 1uF capacitor connected to MP8301.
- Top Right Diagram:** Shows the connection of UG3V3, UG3V3_ETH_1010A, and MP8300. The network includes inductor L8302, capacitors C8313 (22uF) and C8307 (1uF), and a 1uF capacitor connected to MP8300.
- Bottom Left Diagram:** Shows the connection of 1V8_1010PHY and MP8302. The network includes capacitor C8310 (22uF), capacitors C8303 (1uF) and C8308 (1uF), and a 1uF capacitor connected to MP8302.
- Bottom Right Diagram:** Shows the connection of 1V8_1010PHY and MP8303. The network includes capacitor C8311 (22uF), capacitors C8302 (1uF) and C8304 (1uF), and a 1uF capacitor connected to MP8303.

Rev.	type of change	change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved		
				date name	
				created	
		checked			
schematic			Sheet	Sheets of	Sample
4010_ETH_PHY			6	31	
Type			Variant	Part Number	
substitute for:		substitute by:			

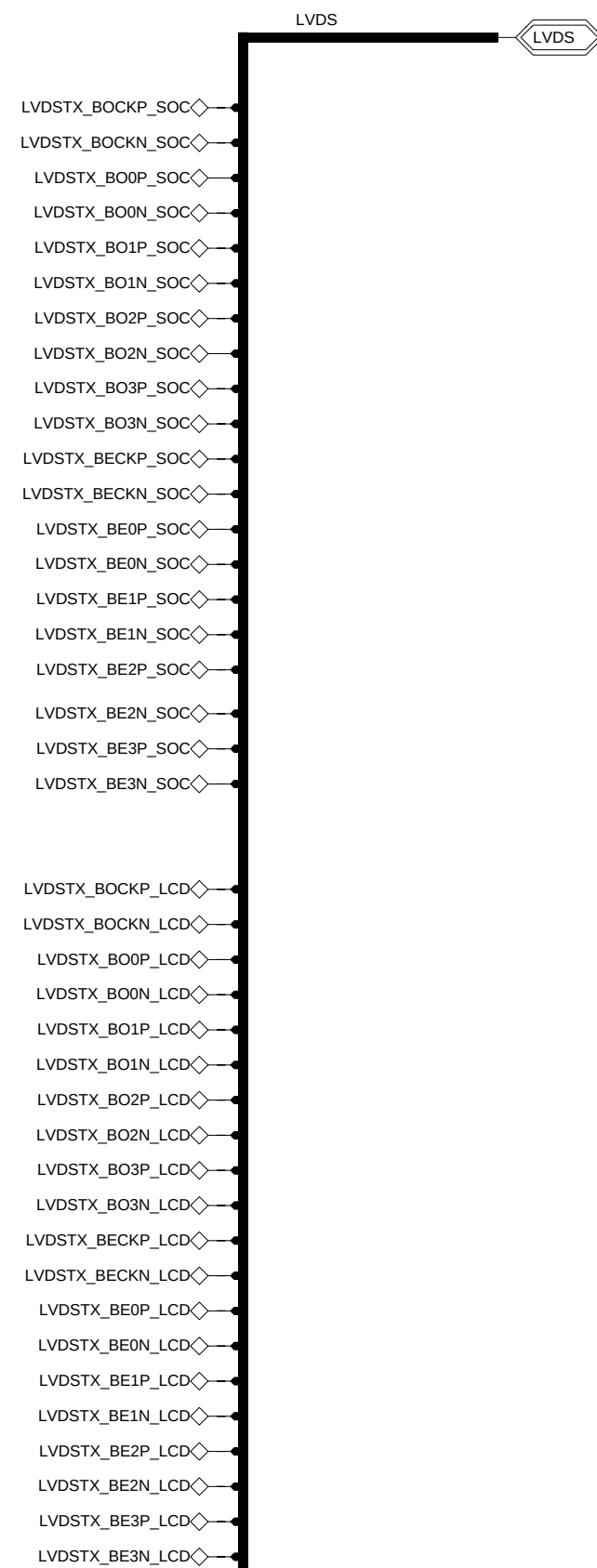
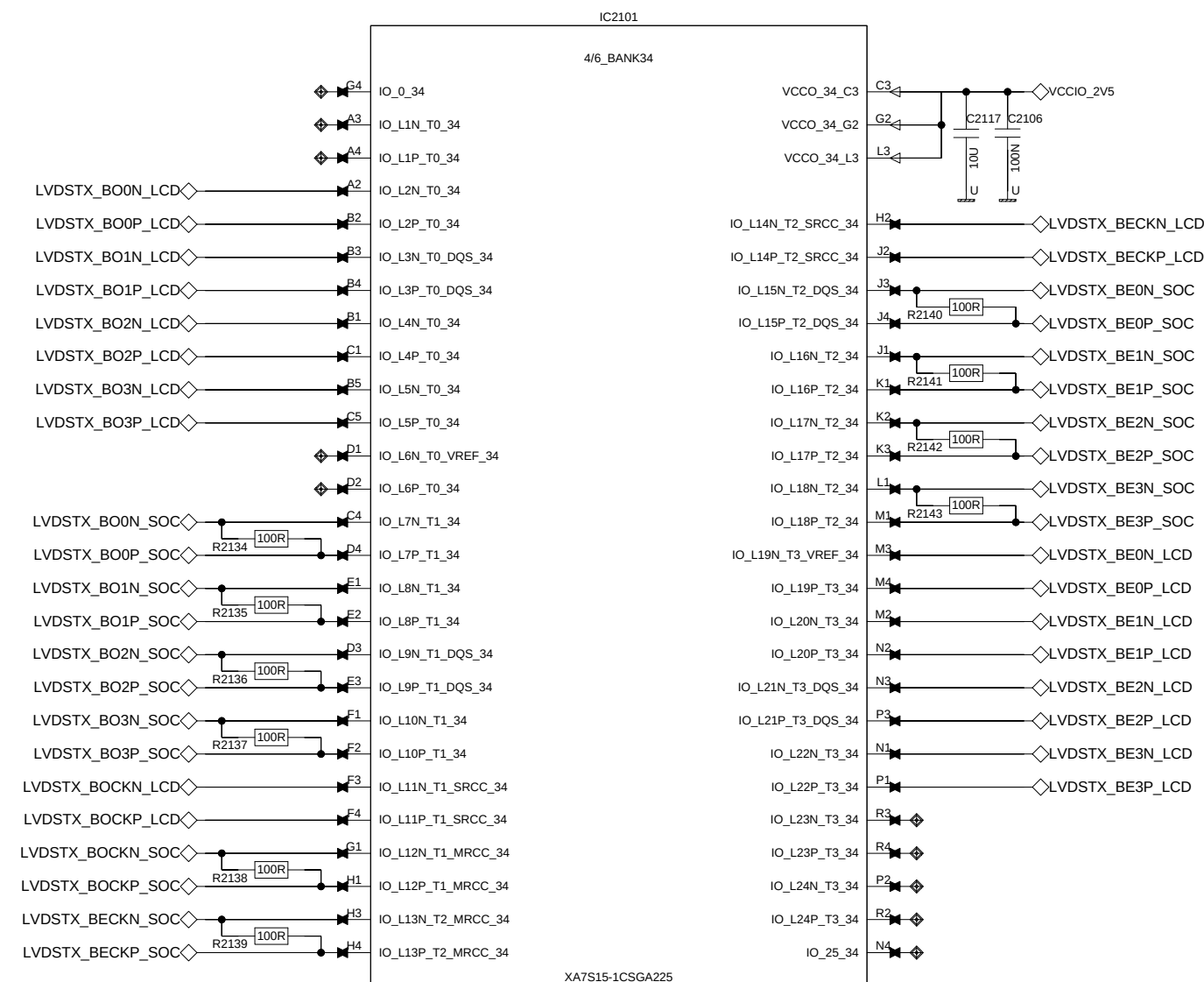
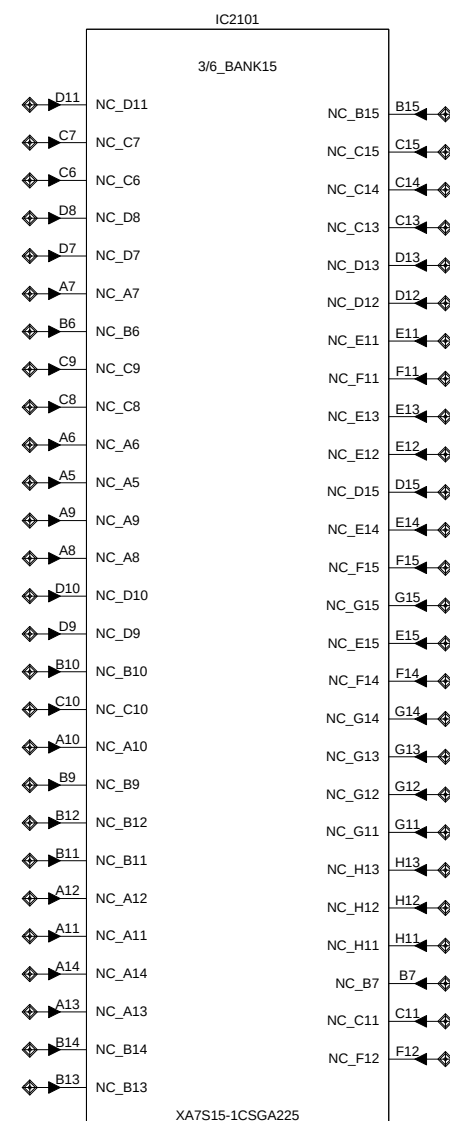
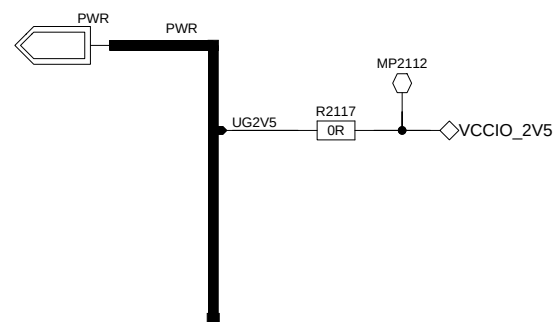




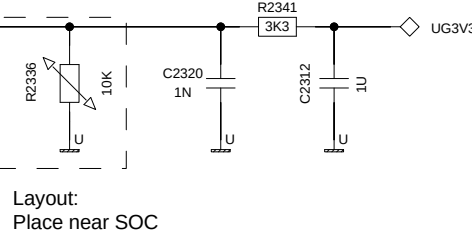
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schematic			Sheet 8	Sheets of 31	Sample
FPGA_CONF			Type	Variant	Part Number
substitute for:		substitute by:		C143	



Rev.	type of change	change-no.	date of change	auth.	checked
	Copyright reserved	date		name	
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schematic	Sheet 9		Sheets of 31		Sample
FPGA_POWER	Type		Variant		Part Number
substitute for:		substitute by:		C143	



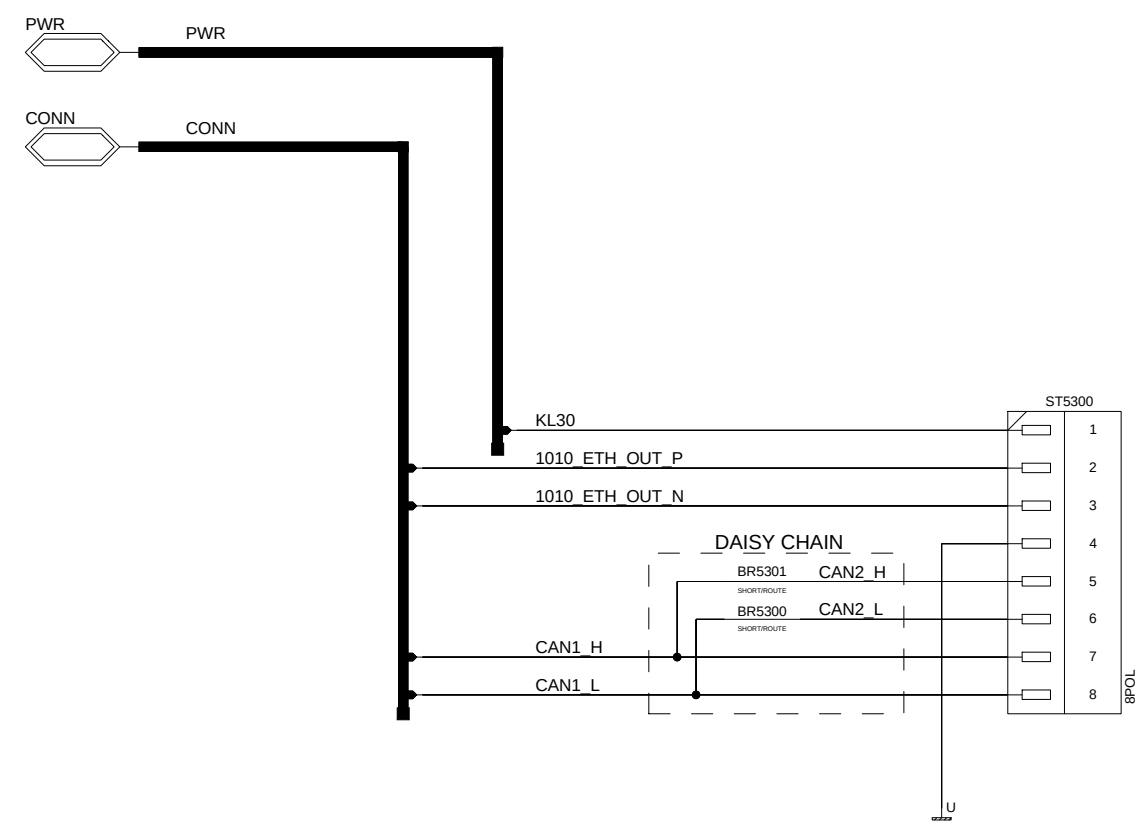
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			checked		
schematic			Sheet	Sheets of	Sample
FPGA_VIDEO			10	31	
			Type	Variant	Part Number
substitute for:		substitute by:		C143	



INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	-----
INTP2	ISO	-----
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	-----
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	-----

Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved						date		name		
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schematic:						Sheet 11	Sheets of 31	Sample		
4010_IDC						Type	Variant C143		Part Number	
substitute for:						substitute by:				

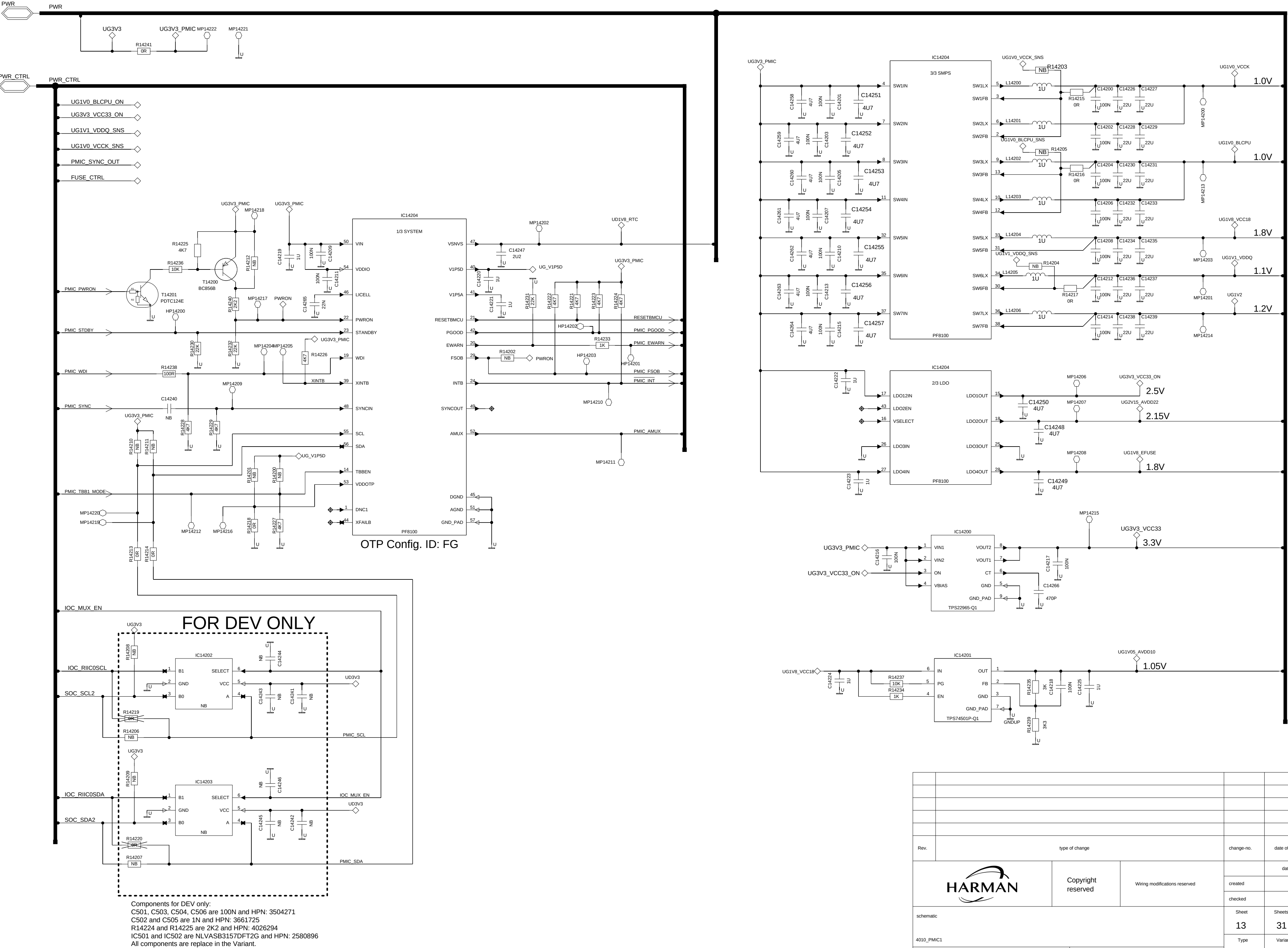
MAIN CONNECTOR (5300-5399)



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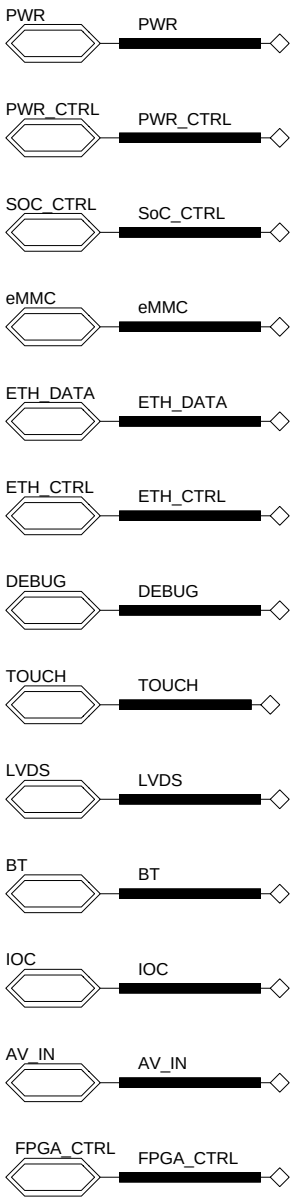
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				checked		
schematic			Sheet	Sheets of		Sample
4010_Main_Connector			12	31		
substitute for:			Type	Variant		Part Number
substitute by:						

PMIC1 (14200-14299)

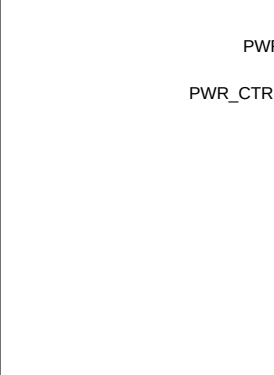


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		Copyright reserved	Wiring modifications reserved	date name	
schematic			Sheet	Sheets of	Sample
4010_PMIC1			13	31	
substitute for:		substitute by:	Type	Variant	Part Number

SOC (10000-10999)

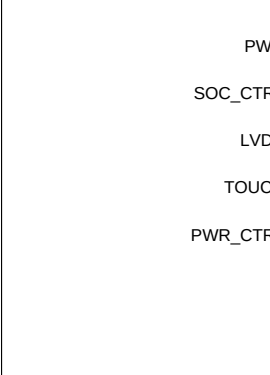


4010 SoC Power



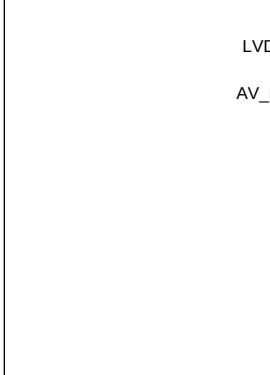
10500 - 10599

4010 SoC_CTRL



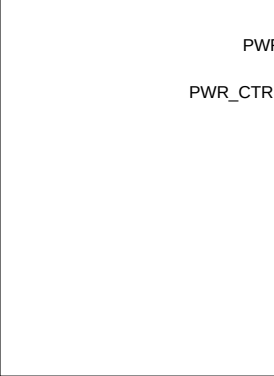
10600 - 10699

4010 SoC Video



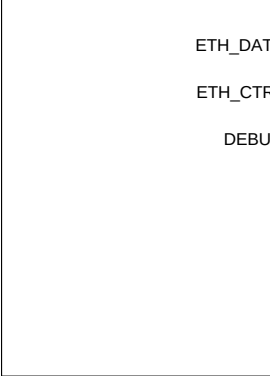
10700 - 10799

4010 SoC LPDDR4



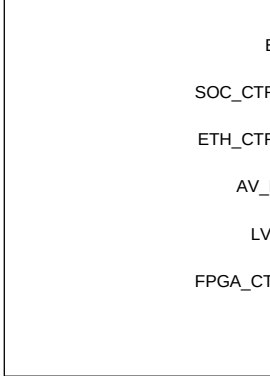
10400 - 10499

4010 SoC_ETH_PCIE_USB



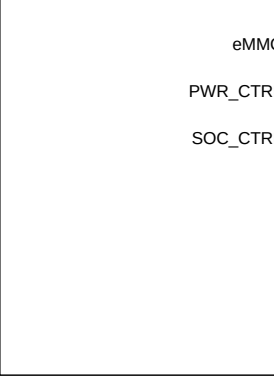
10300 - 10399

4010 SoC Audio



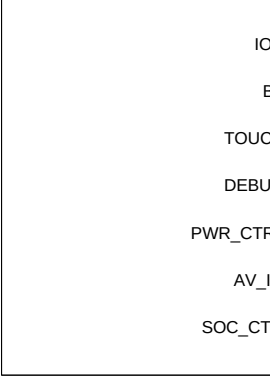
10800 - 10899

4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

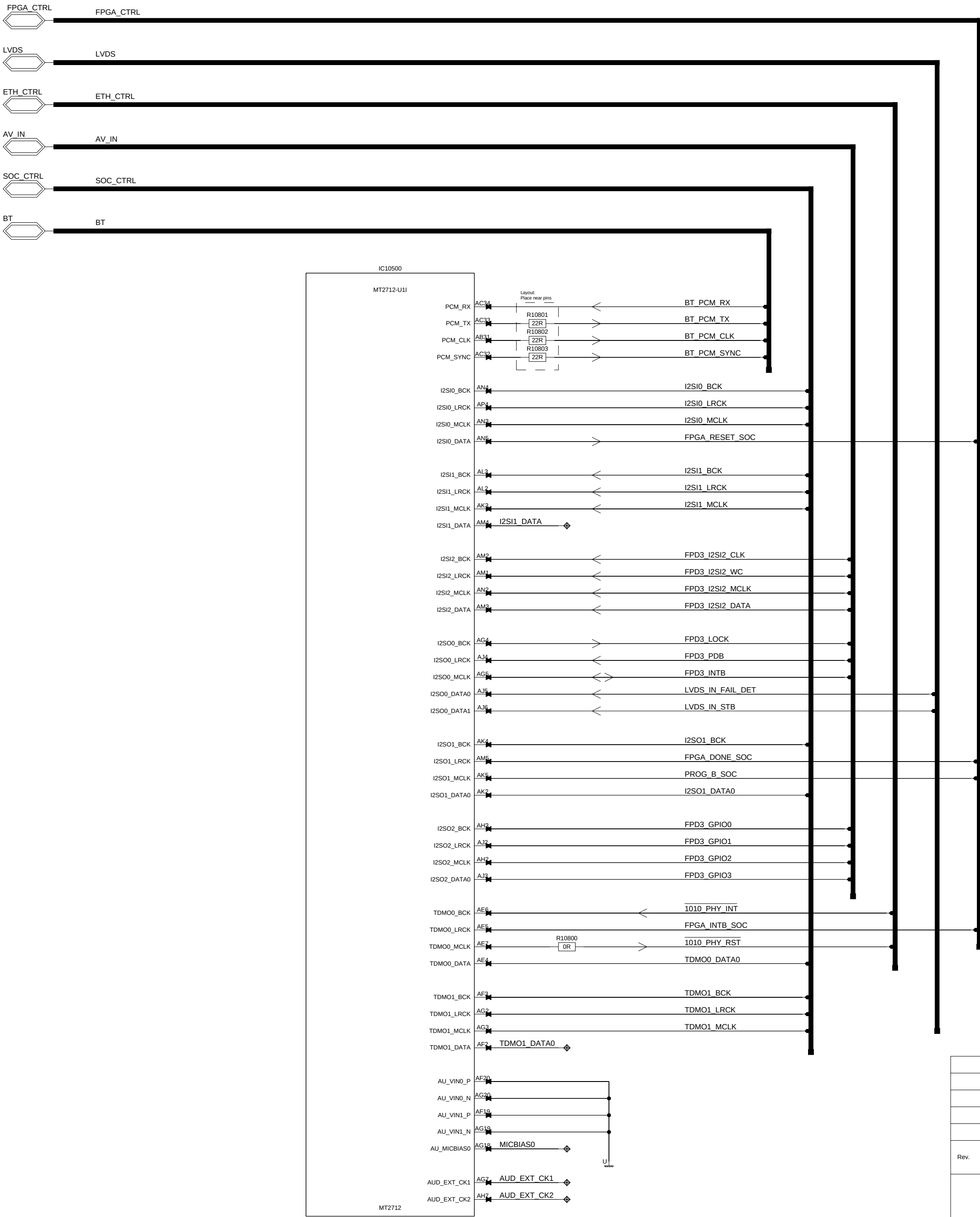


10200 - 10299

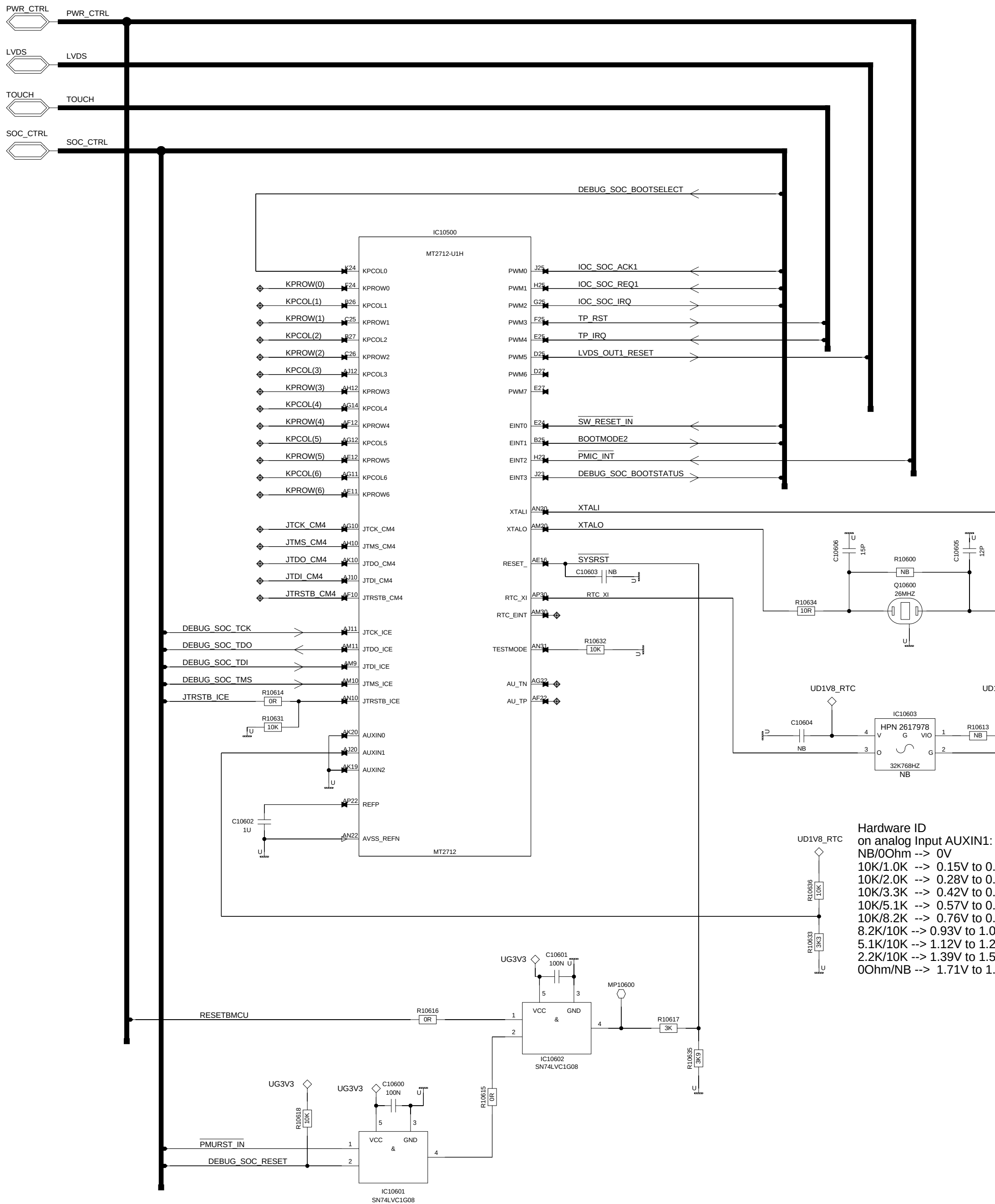
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		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of		Sample
4010_SoC			15	31		
substitute for:			Type	Variant		Part Number
substitute by:						

SOC_AUDIO (10800-10899)



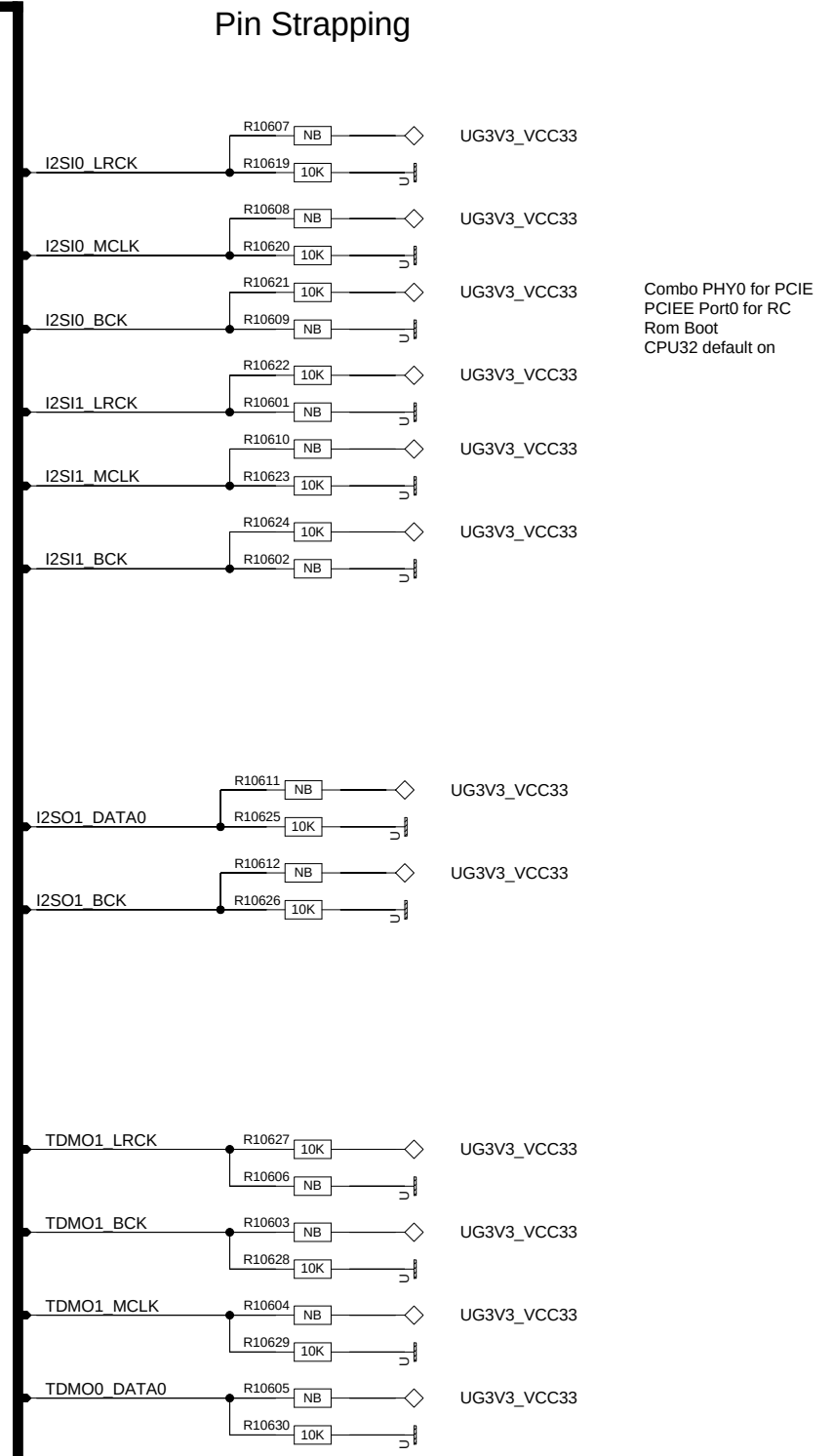
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					created				
					checked				
schematic				Sheet	Sheets of		Sample		
4010_SoC_Audio				16	31				
substitute for:				Type	Variant		Part Number		
substitute by:									



	multi-used				multi-used	
Mode	I2Si0_LRCK	I2Si0_MCLK	I2Si0_BCK	I2Si1_BCK	I2Si1_LRCLK	I2Si1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIE	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIE	X	X	X	X	0	X
PCIEE Port0 for RC	X	X	X	1	X	X
PCIEE Port0 for EP	X	X	X	0	X	X
PCIEE Port1 for RC	X	X	1	X	X	X
PCIEE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

i2S01_DATA0 status[1]	L	L	H	H
i2S01_BCK status[0]	L	H	L	H
1	EMMC	NAND	NOR	SD
2	SD	EMMC	NAND	NOR
3	NOR	SD	EMMC	NAND
4	NAND	NOR	SD	EMMC

Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO1_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

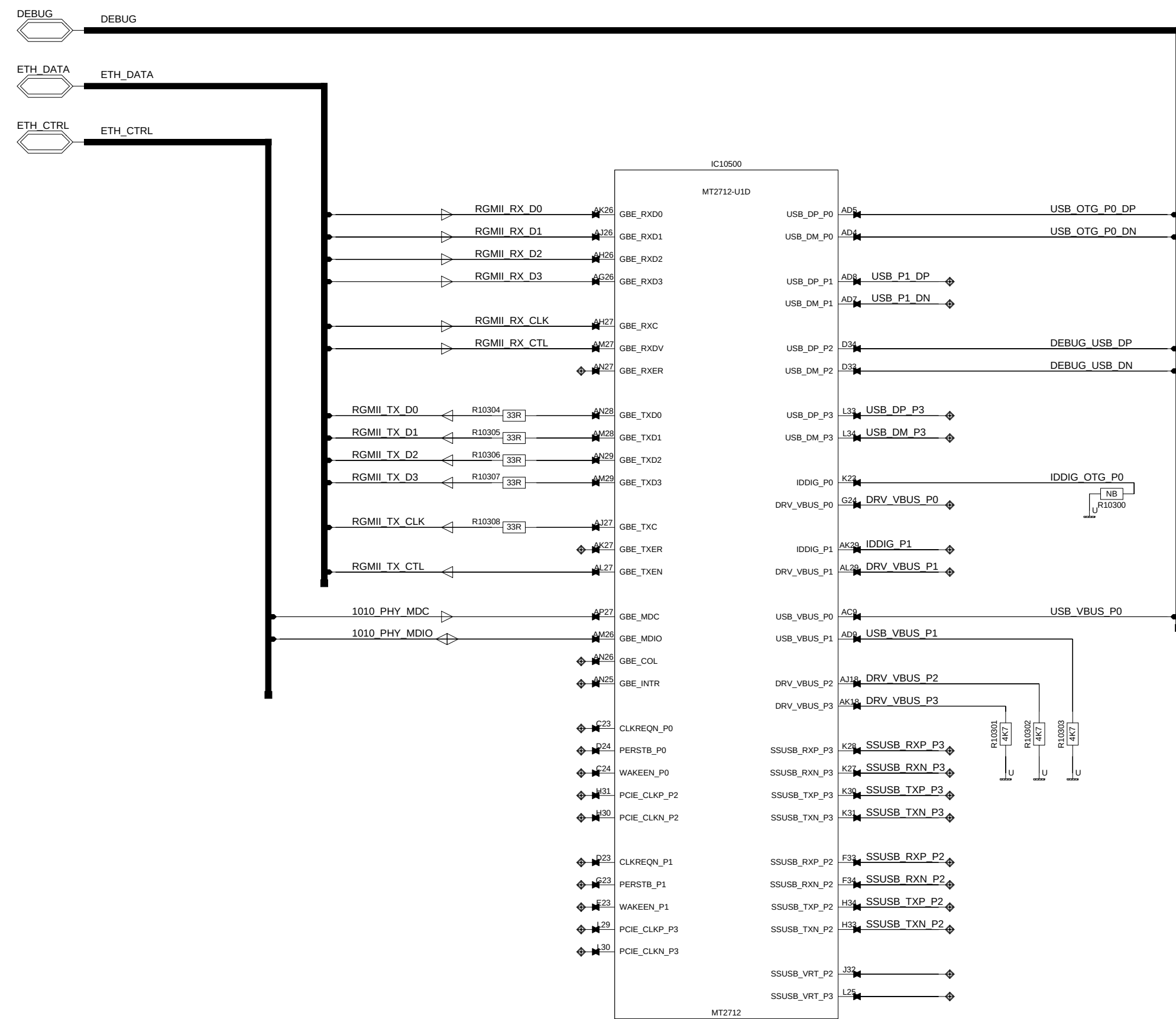


```
Hardware ID
NB/OOHm Input AUXIN1:
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
OOHm/NB --> 1.71V to 1.89V
```

ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

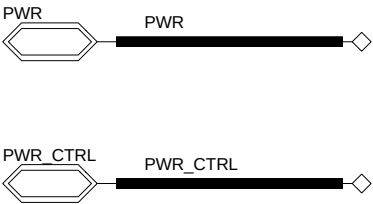
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				created		
				checked		
schematic			Sheet 17	Sheets of 31	Sample	
4010_SoC_CTRL			Type	Variant	Part Number	
substitute for:		substitute by:				

SOC_ETH_PCl_e_USB (10300-10399)

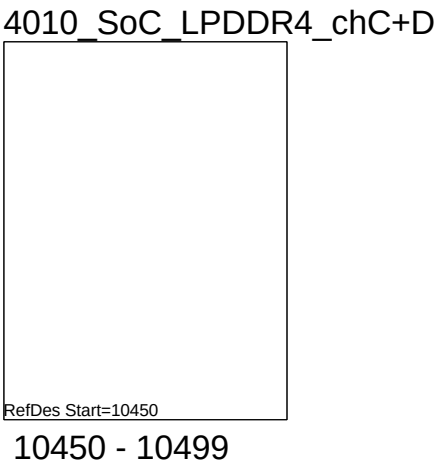
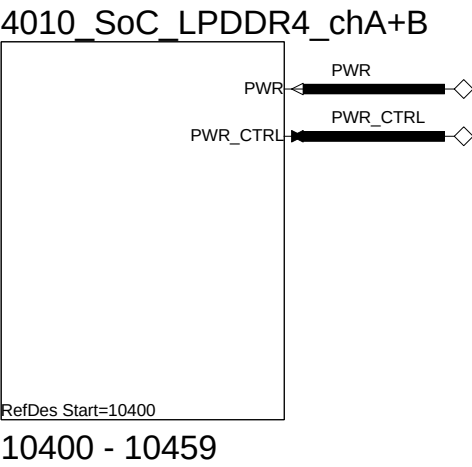


Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
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				checked	
schematic			Sheet 18	Sheets of 31	Sample
4010_Soc_ETH_PCIE_USB			Type	Variant	Part Number
substitute for:		substitute by:			

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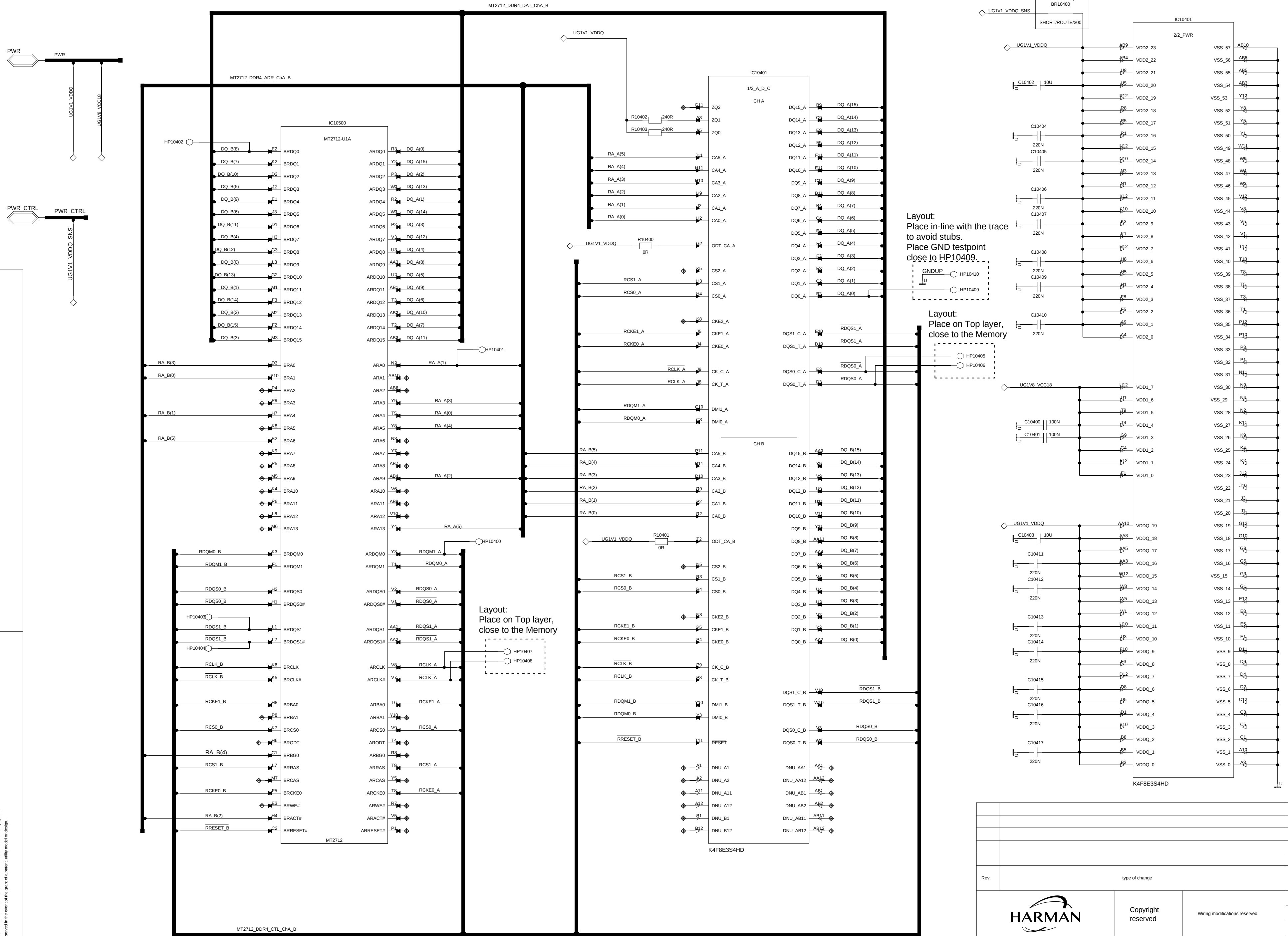


SOC_LPDDR4 (10400-10499)



Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of	Sample	
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant	Part Number	
substitute by:						

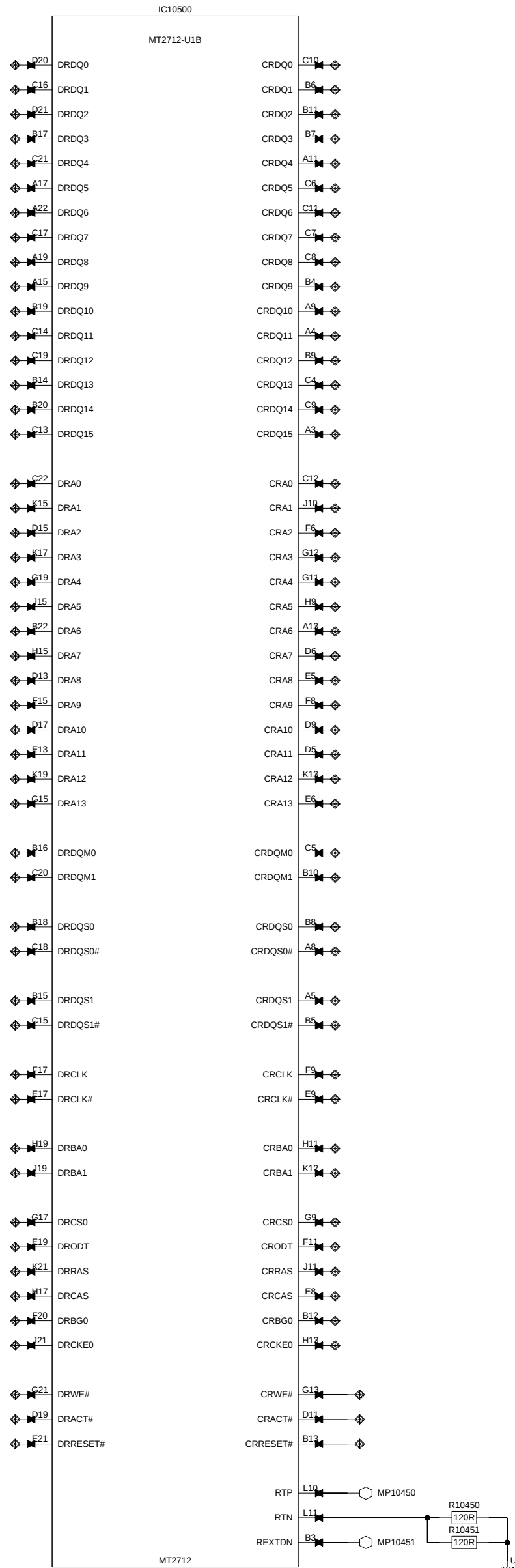
SOC_LPDDR4 (10400-10449)



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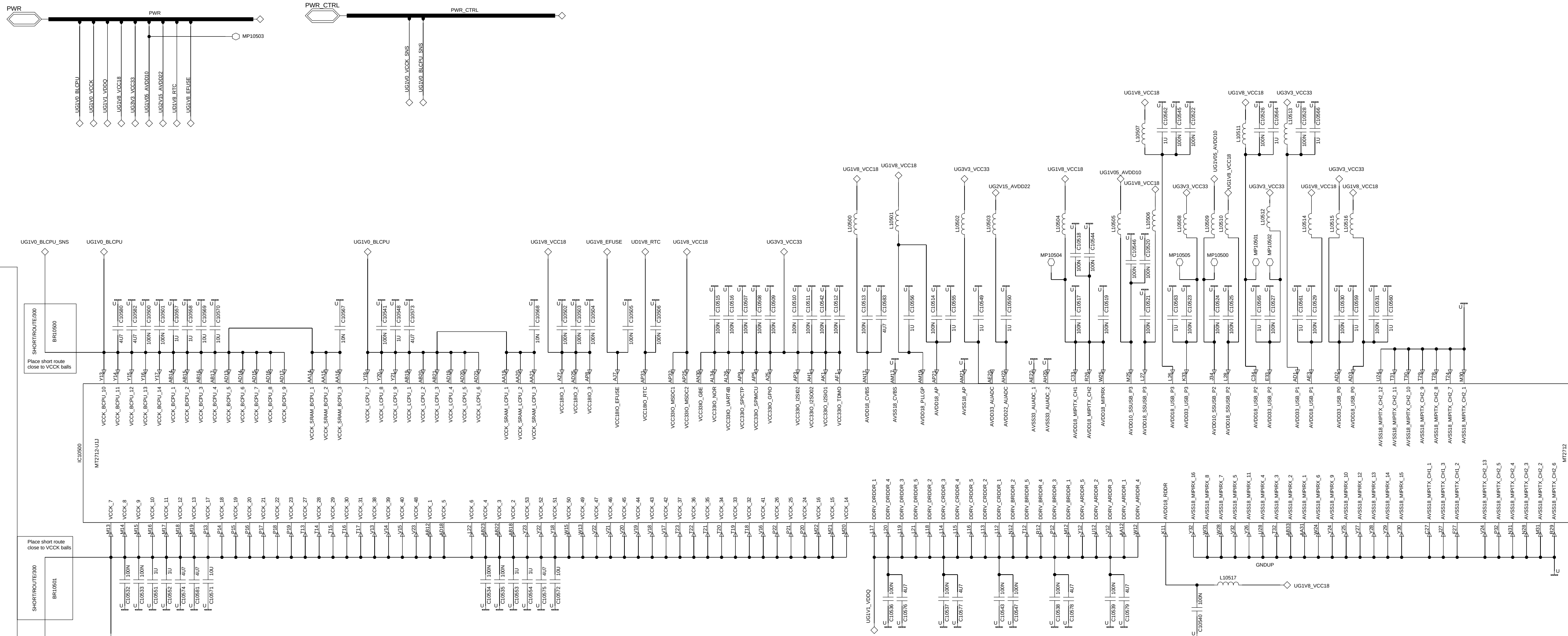
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 Copyright reserved Wiring modifications reserved			date		name
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schematic			Sheet 20	Sheets of 31	Sample
4010_SoC_LPDDR4_chA+B			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_LPDDR4 (10450-10499)

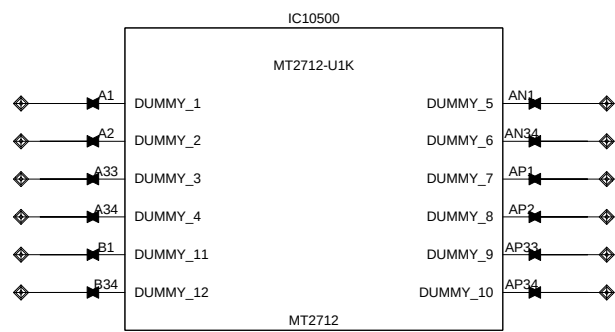


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				created	
				checked	
schematic			Sheet 21	Sheets of 31	Sample
4010_SoC_LPDDR4_chC+D			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_POWER (10500-10599)

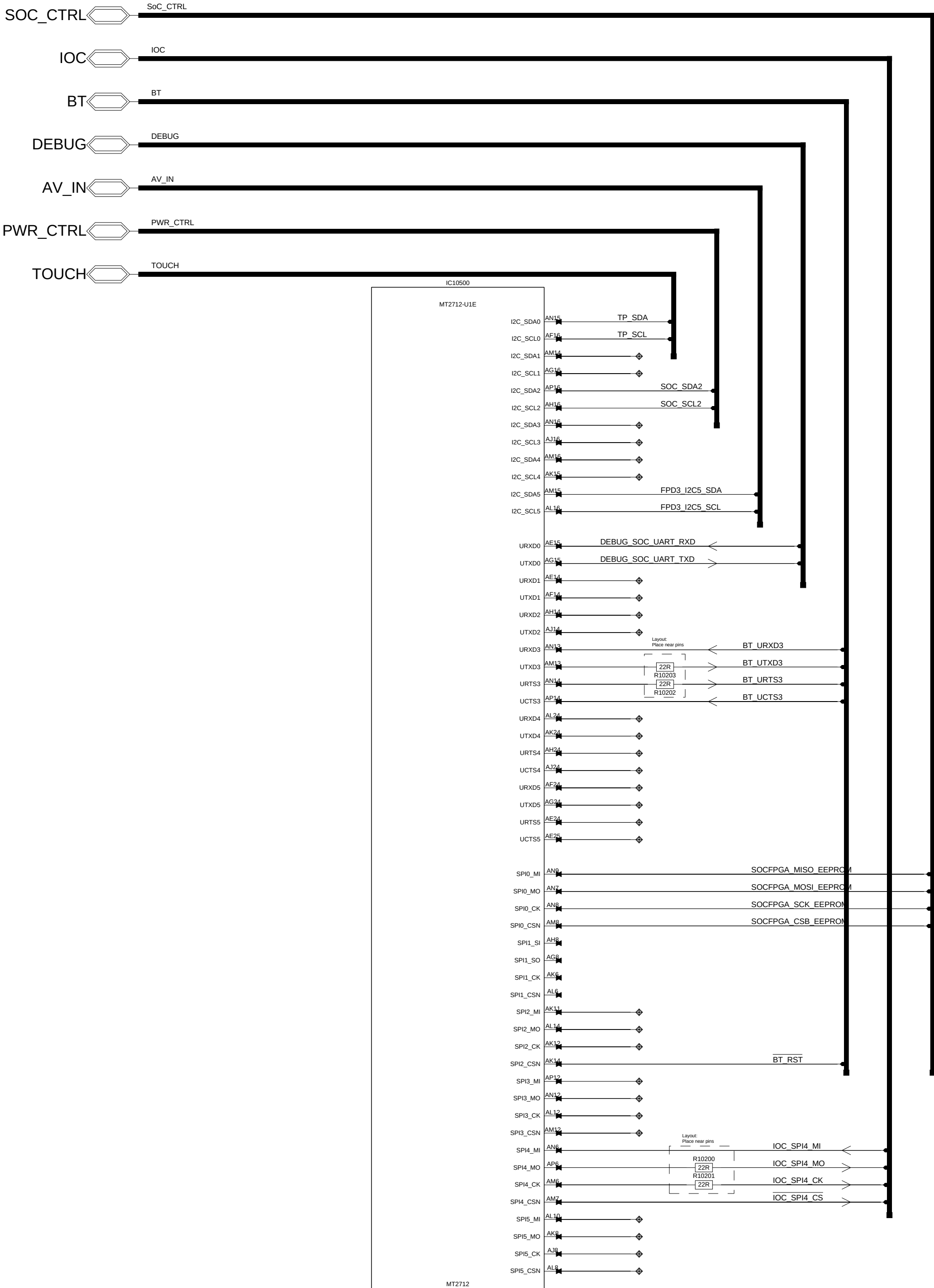


4010 SoC GND



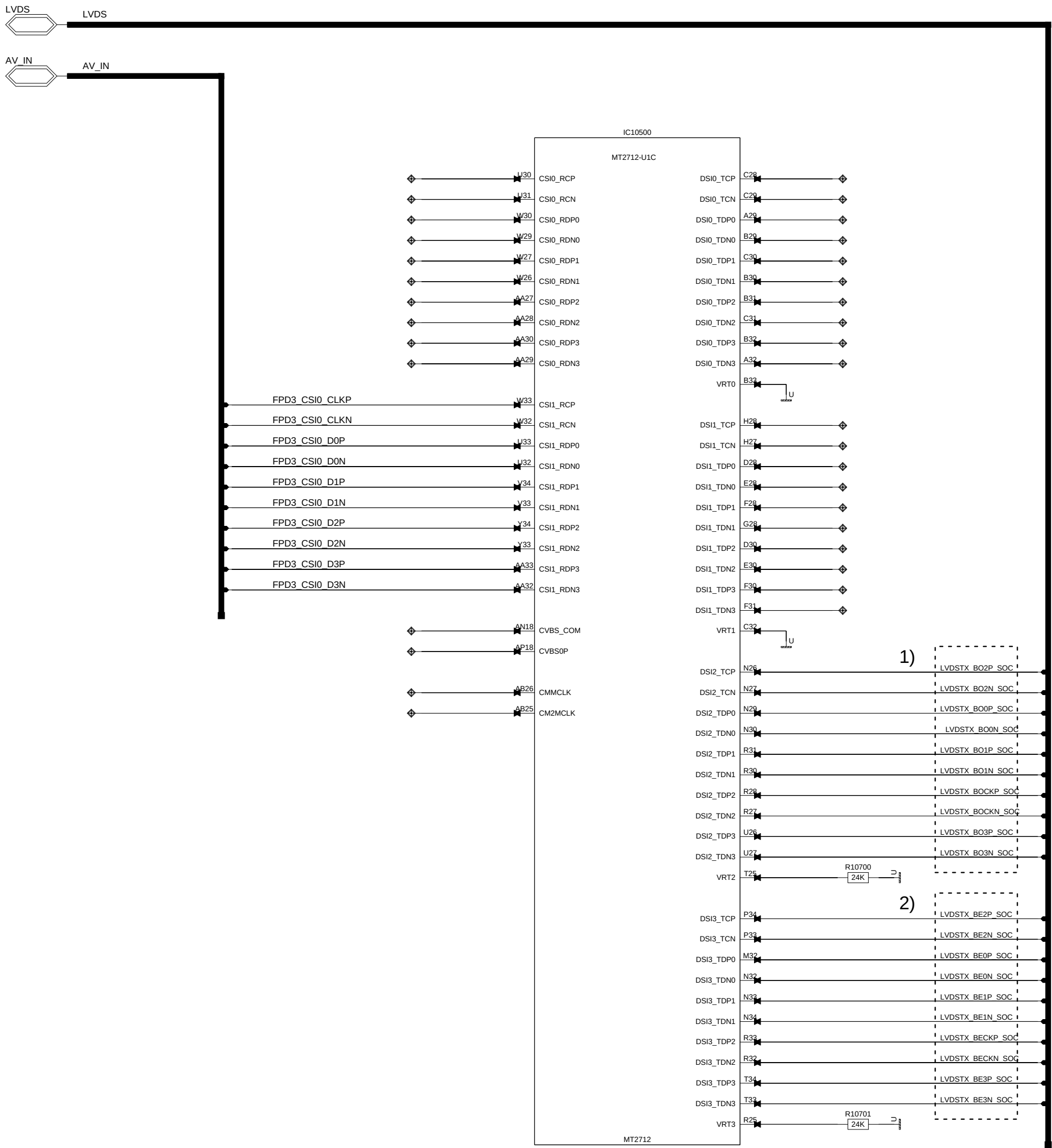
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			created		
			checked		
schematic		Sheet 23	Sheets of 31	Sample	
4010_SoC_GND		Type	Variant	Part Number	
substitute for:		substitute by:			

SOC_SERIAL_IF (10200-10299)



Rev.	type of change		change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic			Sheet	Sheets of	Sample	
4010_SoC_Serial_IF			24	31		
substitute for:			Type	Variant	Part Number	
substitute by:						

SOC_VIDEO (10700-10799)



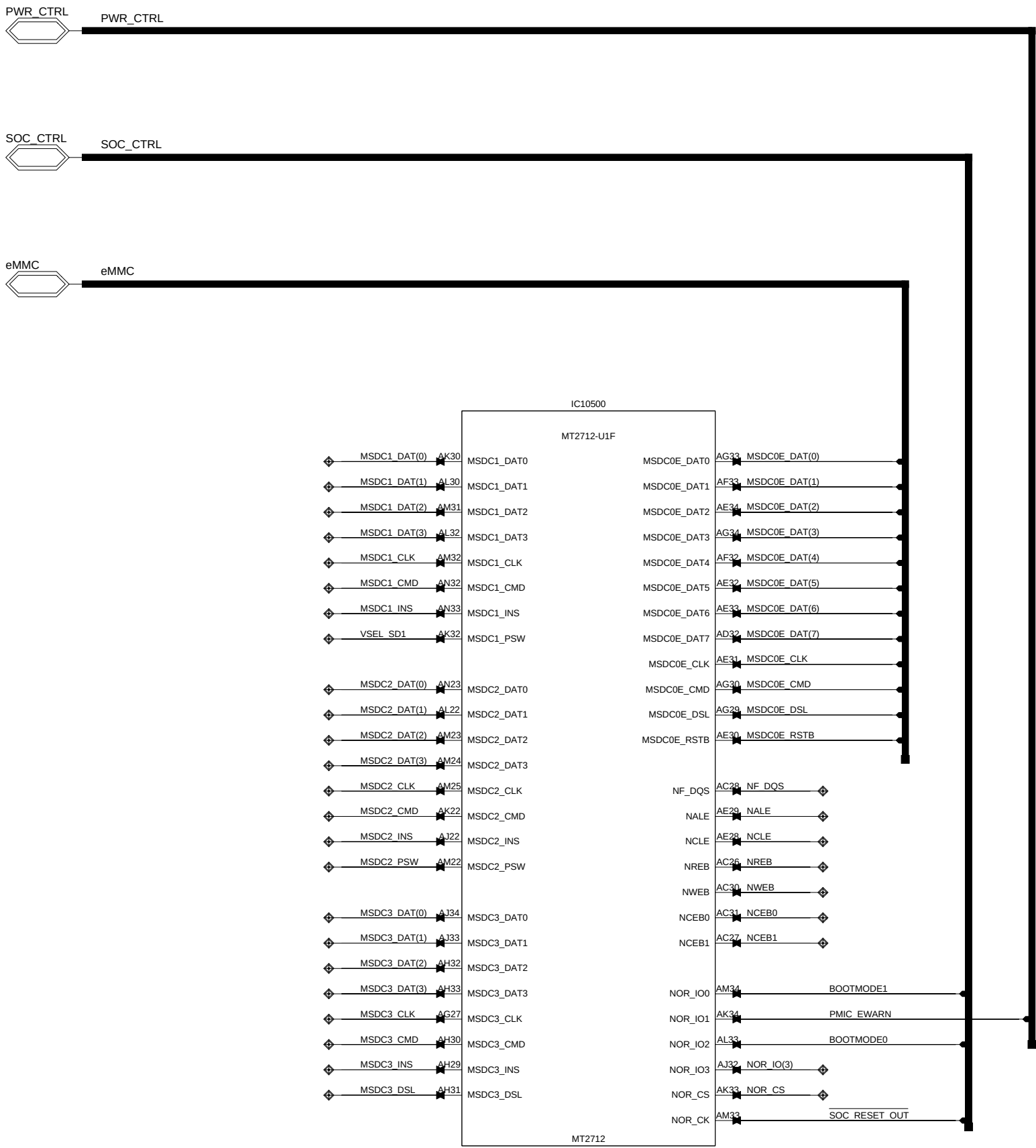
- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BOCK) to data (BO[0-3]): 5mm

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic 4010_Soc_Video			Sheet 25	Sheets of 31	Sample
			Type	Variant	Part Number
substitute for:		substitute by:			

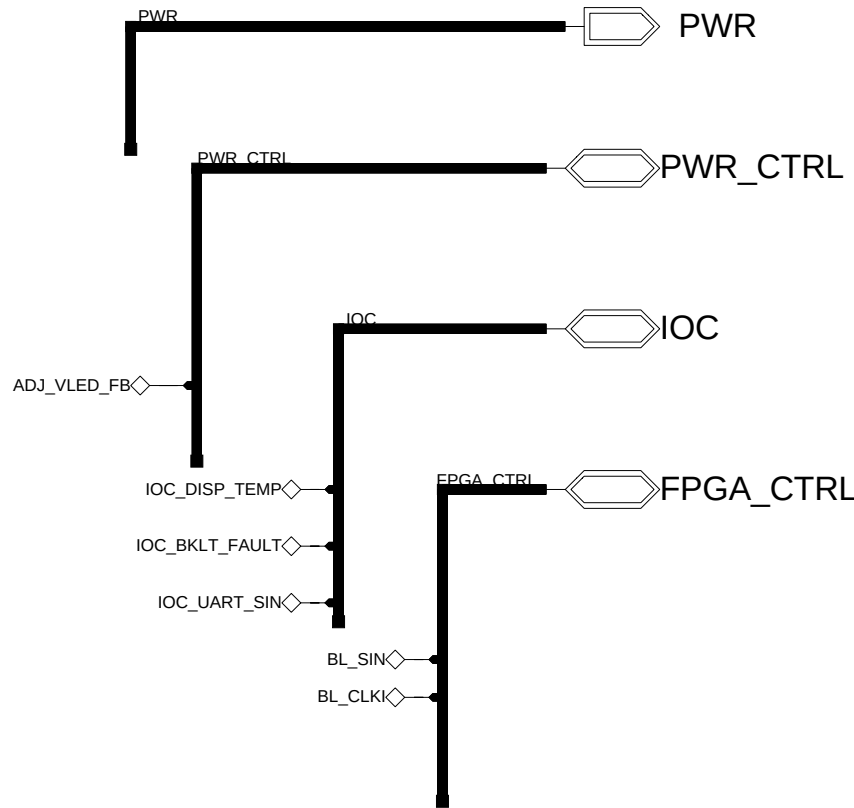
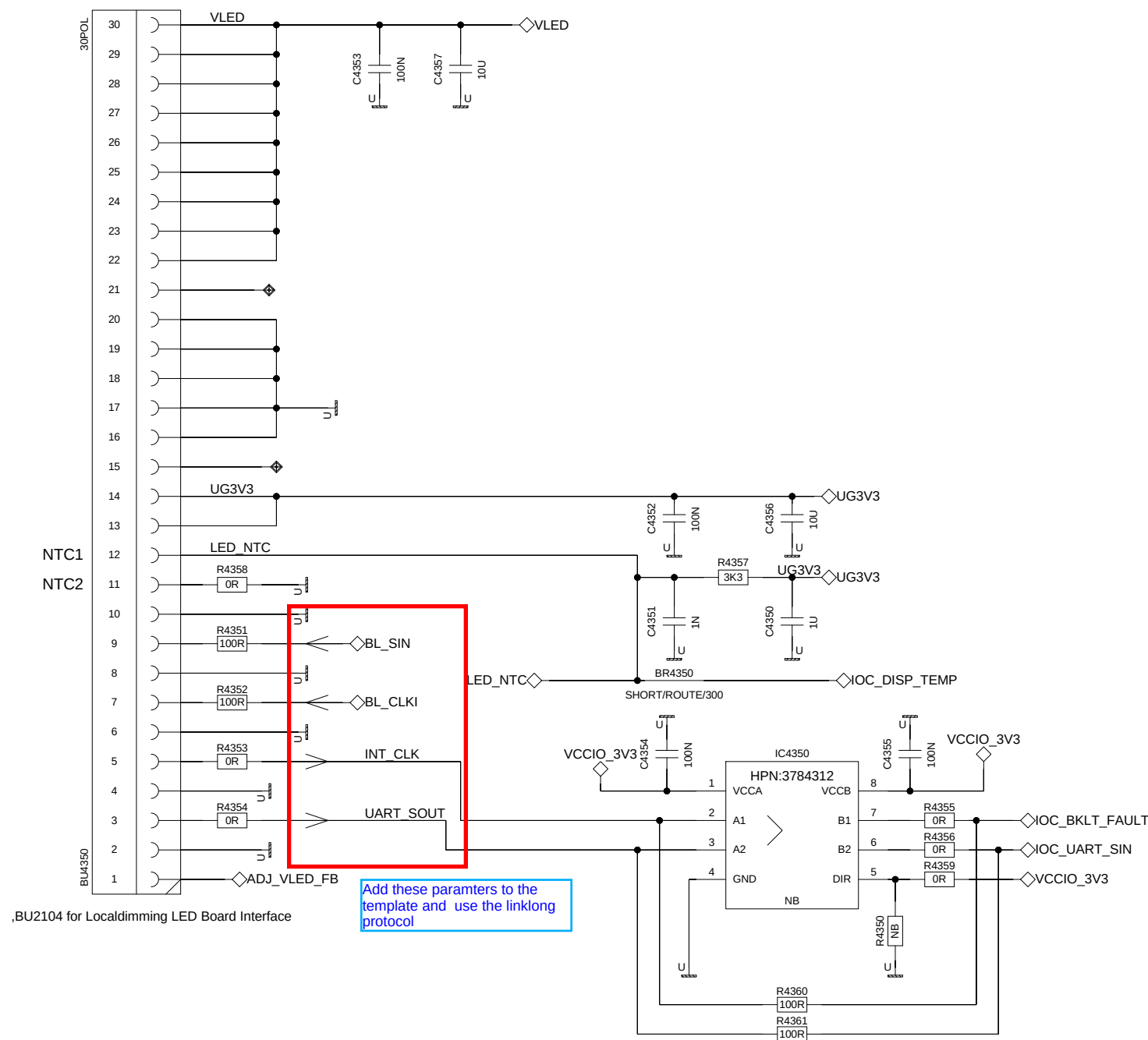
A3 04.02.12

SOC_eMMC_NAND_SID (10100-10199)

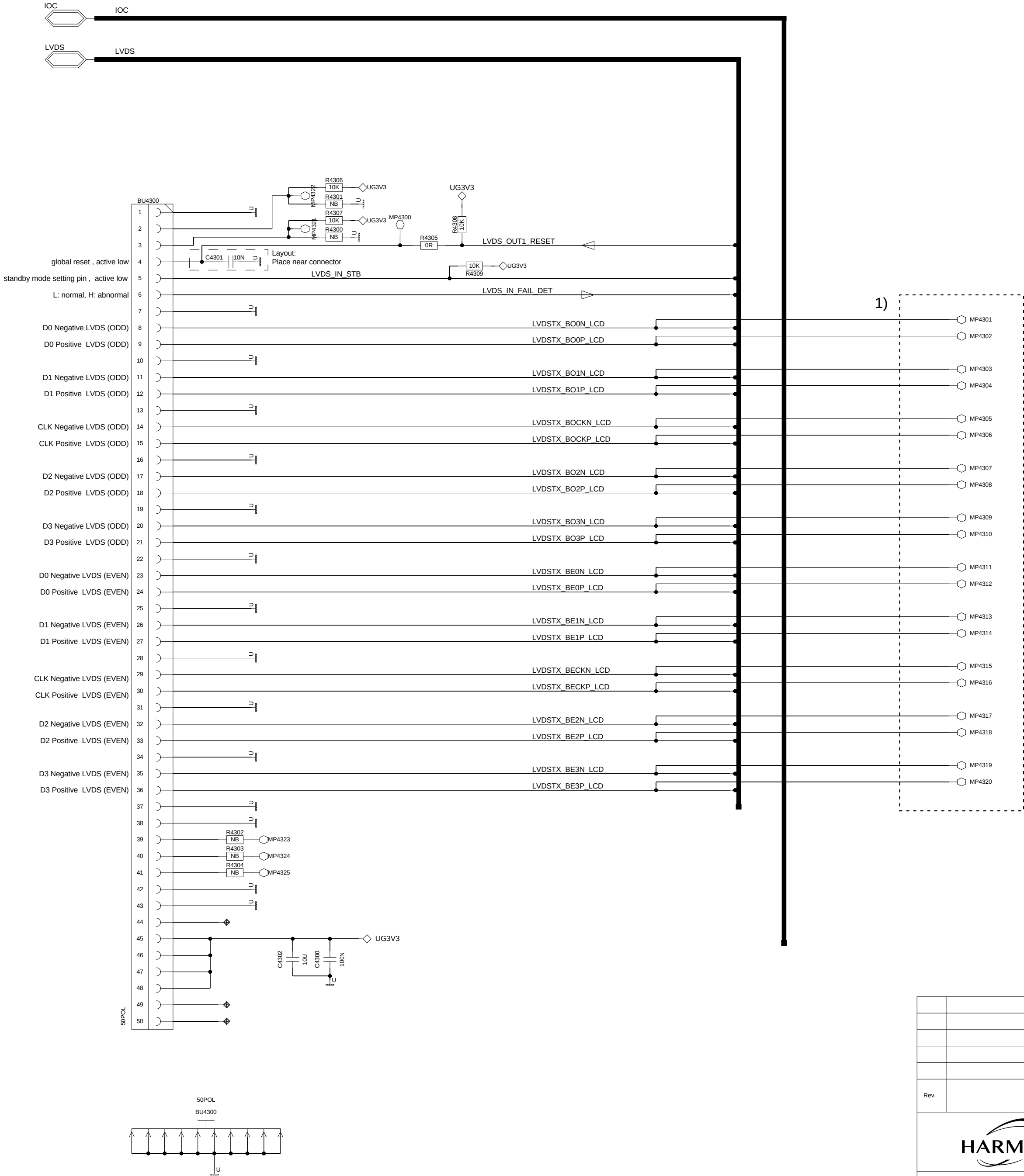
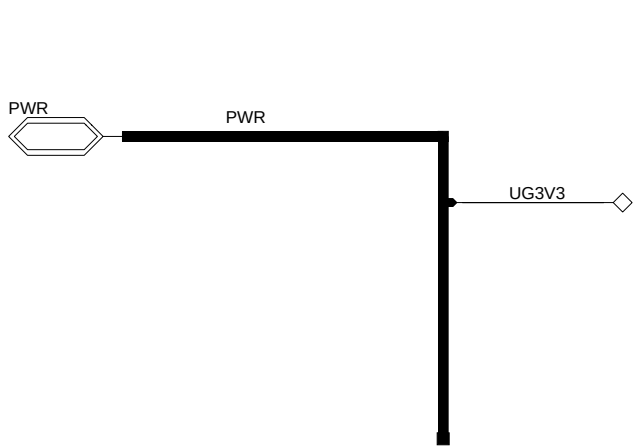


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		Copyright reserved	Wiring modifications reserved	date		name
				created		
				checked		
schematic				Sheet 26	Sheets of 31	Sample
4010_SoC_eMMC_NAND_SD				Type	Variant	Part Number
substitute for:		substitute by:				

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				checked	
schematic			Sheet 28	Sheets of 31	Sample
4010_TFT_Bl_Conn			Type	Variant	Part Number
substitute for:		substitute by:		C143	

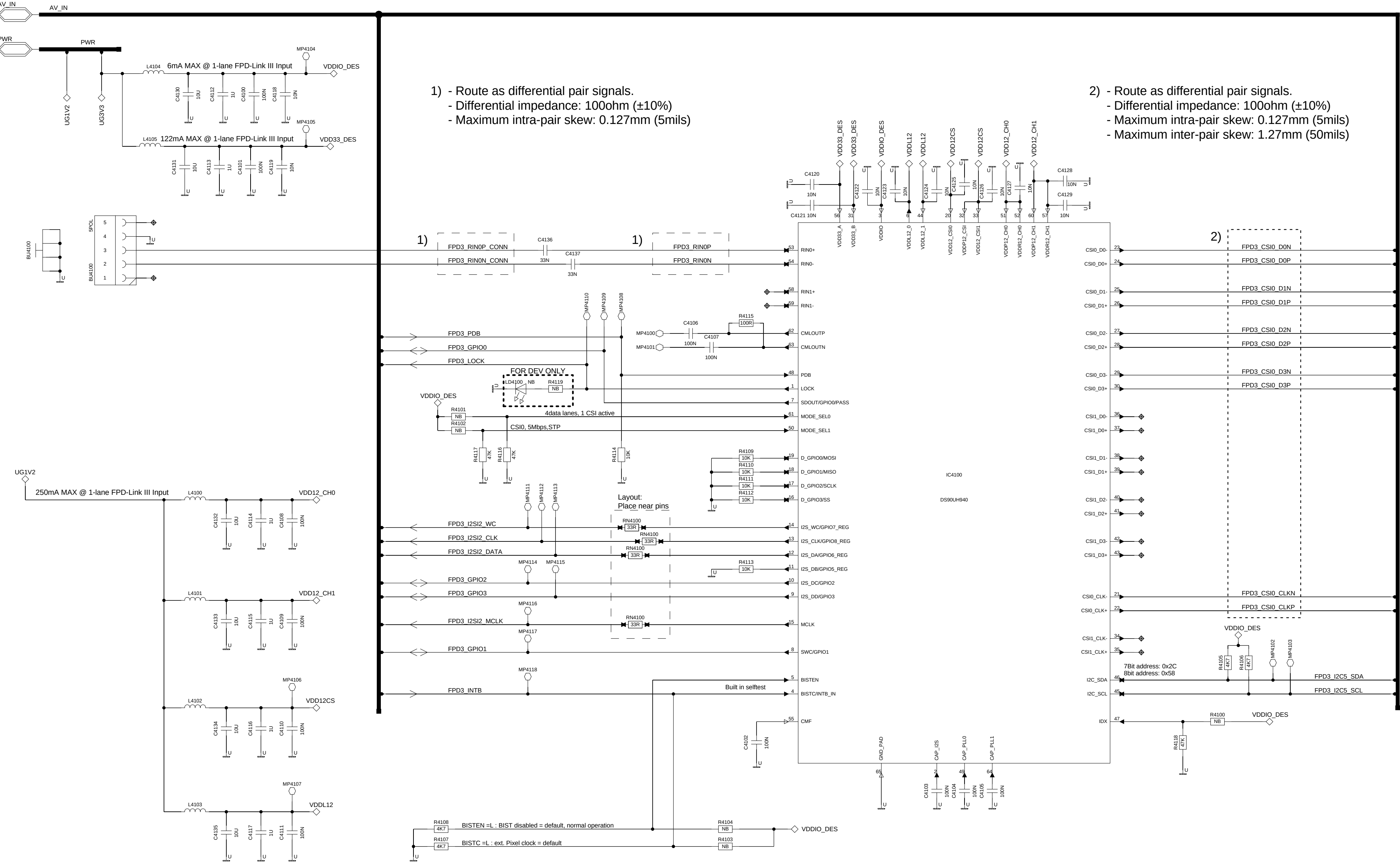


Verify pin out during layout

1) Testpoints should be placed as close as possible to an existing VIA

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			checked		
schematic			Sheet	Sheets of	Sample
4010_TFT_Conn			29	31	
substitute for:			Type	Variant	Part Number
				C143	
substitute by:					

VIDEO_DES (4100-4199)



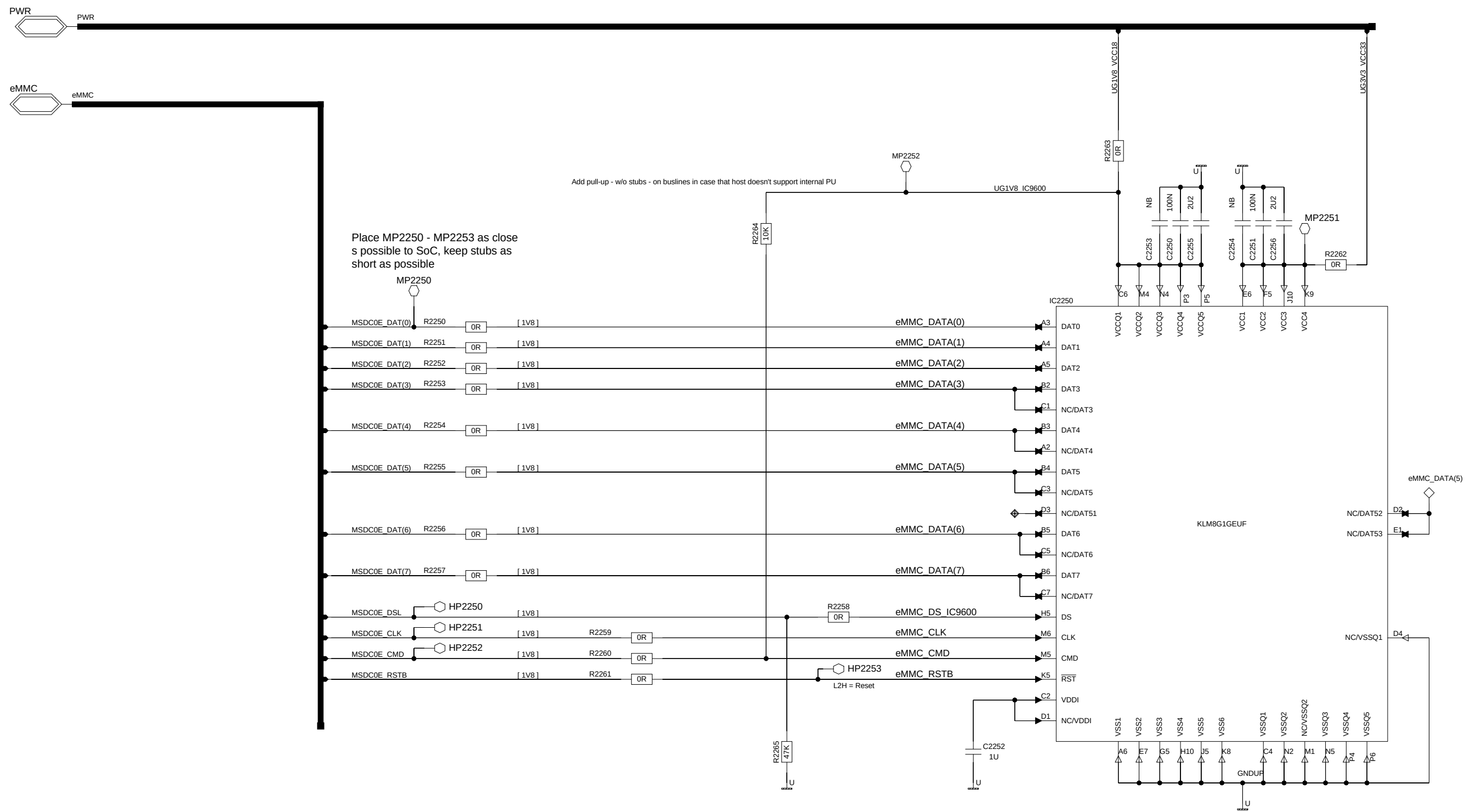
- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum inter-pair skew: 1.27mm (50mils)

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		Copyright reserved	Wiring modifications reserved	date name		
				created		
				checked		
schematic				Sheet 30	Sheets of 31	Sample
4010_VIDEO_DES				Type	Variant	Part Number
substitute for:		substitute by:		C143		

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617